

Automated Course Seeking AI-Chatbot using Natural Language Processing (NLP)

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ABSTRACT

Nowadays, many people purchase courses online to upskill themselves for own interests or work purposes. Choice of course available are plenty but many are costly to get the certification. This will result in people who is poor are not able to learn through online course. This study is to examine the learning preferences and financial status of people aged 18 and above and create a Chatbot to learn their preferences and recommend the desired course for the users. The researchers have collected 41 survey responses from people aged 18 and above to justify the problem statement that the researcher provided is valid by using qualitative analysis method. Analysis is done to produce the requirements of the proposed system. A processed dataset from Kaggle is used as a data source for the chatbot and the implementation of the chatbot on a website. This project uses Rapid Unified Process (RUP) methodology to build the chatbot using the python natural language toolkit. This project is to provide solution to the SDG 8 to provide sustainable and decent work for all humans.

Keywords: Natural Language Processing (NLP), Chatbots, Machine Learning, Course Seeking System, Online Learning, Artificial Intelligence (AI)

SDG Goal 8: Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all

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CHAPTER 1: INTRODUCTION

1.1 Introduction

In the modern era, online learning is a new trend because knowledge and information are easily searchable through the internet. They may come in different virtual forms: traditionally, such as digital news, news streaming platforms, articles, digital journeys, and research articles, to less formal types of information acquisition, such as TikTok and YouTube videos, and educational creators on Instagram. People can gain knowledge with a single touch of their fingers on smartphones. However, variations in the quantity of knowledge on the internet could raise many issues related to the quality, relevance, and integrity of the information when one is serious about learning a skill set.

Learners have other plans before continuing their studies after completing a phase of academic milestones such as College or Secondary. Regardless, they need to explore their interests by gaining knowledge in different fields to see which one best suits them. Some common methods are through YouTube videos or purchasing courses to try them out. However, some tutorials are too lengthy, have bad quality, outdated materials or provide outdated content, they cause learners to waste lots of time and money.

Artificial Intelligence (AI) has been around for quite some time. According to Copeland (2023), AI is the ability of digital computers or robots to perform tasks commonly associated with intelligent beings. Most modern AI chatbots can comprehend human input and reply in a natural way of speaking. This is achieved using the technology of Natural Language Processing (NLP), considered a branch of AI that helps computers understand, interpret, and manipulate human language. (Natural Language Processing (NLP): What It Is and Why It Matters, n.d.) AI chatbots with NLP can provide decent responses depending on user input and perform some actions using predefined task automation. Automation refers to automating workflows on the computer as well as using online tools to execute instant commands. AI chatbots with automation are very convenient and powerful tools for helping with daily human tasks.

As the rise of Artificial Intelligence adopts the use of natural language processing (NLP), powerful AI tools are emerging to help humans in their daily lives by utilizing NLP and training models with big data techniques to acquire more accurate data in a more human-like manner, as seen in tools like ChatGPT and Barn by Google. Recently, it has become a norm for society to use AI after the launch of ChatGPT by OpenAI on November 30, 2022 (OpenAI, 2022), as a helper in their work, providing guidance and ideas after receiving prompts from users.

1.2 Problem Context

After an introduction to the background of the field of online platform and course-seeking tools, there are several problems has been encountered.

The first problem is there is an overwhelming number of courses available on those platforms which might cause difficulty in choosing the suitable course for them. The quality of the courses on the platform varies which will cause the problem stated. The reason behind this is there are courses related to one topic but they are from different creators, different levels of tutorials as well as teaching styles and lengths of tutorials. That will be time-wasting and money-wasting for those who just jump into a course which might result in relevance for the topic but the contents are not relevant at all or do not fit the level of knowledge for that user. For people who have a limited budget for learning, free tutorials are the best options for them. However, according to research by Clifton & Mann (2011), content on free sources such as YouTube tends to be misleading, inaccurate, and biased. Learners need to have the ability to identify which and where to find the best learning courses on the internet.

Many learners, particularly pre-university students, often have little or no ideas about what they want to learn during their exploration. This commonly occurs for secondary fresh graduates who have to decide their academic path or jump straight into the workforce. This is worsen when it comes to paid courses and videos. For those who directly enter college life, whether compelled or out of curiosity, standard courses in colleges and universities can take 3 to 4 years and cost hundreds of thousands of ringgits to complete a degree certification course. If students realize this is not the path they desire, they face the tough situations of either persisting through a stressful experience or dropping out midway. This resulted in wasting both money and several years in what people called an exploration period. Many struggles in their further studies due to a lack of guidance and help.

Furthermore, there are lack of tools that can recommend a complete and comprehensive path for the learner to get a professional certification. Most of the learners will go to the website that provides the professional certification program, but they will end up with only the list of things they will be learning, exams to participate in, pricing as well and what they can do after completing. However, the pre-requisite and the required knowledge are not stated on the website which causes the learners not to know how to prepare for those certification programs. They need additional searching through the internet to help or to seek help from their seniors who have experienced. Some certifications are outdated but not stated on the website.

1.3 Project Aim

The project aims to develop an AI chatbot using natural language processing technology that can help students find the most suitable courses depending on their needs and budget on online learning platforms and provide a complete path for professional certification.

1.4 Objectives

The project's objectives are as follows:

- To explore the fundamentals of natural language processing with AI chatbots and their implementation on the market.
- To explore similar systems using AI chatbots served as an advisory tool for users in the past.
- To develop an AI chatbot using natural language processing to personalize the learning experience.
- To evaluate the chatbot's usability in realizing an effective learning experience for learners.

1.5 Scope

The researcher planned to use Python to build an AI-chatbot using natural language processing with a trained model with datasets about course learning. The system is equipped with a chatbot that helps the users to solve their problems. The chatbot will also navigate to the respective page upon user request. The chatbot will be able to speak and ask questions to get the user preferences to produce the desired outcome. Each of the chatbots can support one user session but cannot save the chat history and bring it to the next session when the same user is using the bot. This system is a desktop application that is installable on Windows 11.

The chatbot is an advisory tool for learners to find the best-suited course for them. The chatbot is not knowledgeable enough to answer question that is not related to course information and seeking. Other than that, it only can answer using the trained data provided by the researcher. The data might not update to the present day but the latest date that the researcher can find. It does not support on mobile platform. Other than that, it does not support chat history exporting and saving. Only the English language is supported by the chatbot.

1.6 Potential Benefit

Tangible Benefits

This system is aimed at helping learners to find the best-suited courses for them in a small amount of time. All courses are being analysed and recommended for the users according to their needs which will increase the exposure for each course. This can be achieved by putting the latest and also new courses to new courses to be exposed. It can also save costs for the platforms to advertise their course. This will give the learners to have job opportunities after completion of their course or certification.

Intangible Benefits

This system helps in enhancing user experiences on courses seeking to avoid frustration and loss when they find an opportunity to upskill themselves. It provides convenience for the user to compare and find the best course for them from different learning platforms. It will also provide a streamlined complete career path for the learners who want to get certification and the links to the resources. It will provide comfort for users to exchange ideas on career paths with other users on the system by exchanging experiences and notes.

Target User:

This system with a chatbot is built for all the learners who are looking for a tool to help in finding the best course for them to upskill or get certified for themselves.

1.7 Overview of the IR

CHAPTER 1: INTRODUCTION

This chapter gives a brief introduction to AI-chatbots and the development of natural language processing (NLP). The problem background is related to users' browsing experiences on online learning platforms and the issue related to the course-seeking tools available on the platform. Project aim, objectives and scope are included in this section with justifications. The potential benefits party of this project as well as the list of tangible and intangible benefits brought by this project is listed. An overview of this project is included and finally, a project planning for the whole investigation report is included.

CHAPTER 2: LITERATURE REVIEW

In this chapter, the researchers do reviews on pieces of literature and articles that are related to the domains related to this title. This research and investigation are important to support the proposed problem statement and the need to develop the proposed system at the end of this project. The literature review gives a summary of the latest domain research works that can give a brief idea of the current development and status of the technology used in this project. Methods of data collection and analysis, project development techniques, and technology used are investigated and justified by comparing with other alternatives available. This chapter includes domain research, similar system research, technical research and summary.

CHAPTER 3: METHODOLOGY

The chosen research method for system development, data collection and analysis is discussed and justified in this section. This is to ensure the methods used for the research process are feasible and most suitable for this project which can help to improve the quality of the project research result.

CHAPTER 4: CONCLUSION

In the last chapter, the conclusion will provide a brief discussion of the achievements from the first part of the project and justification on the investigation is adequate for the report. Gaps in the research and design will be stated for the researchers’ records as they could refer the possible future enhancements for the project.

1.8 Project Plan

Name of Tasks	Durations	Start Date	End Date	Status
Project Proposal Form	14 days	Tue 5/9/23	Fri 22/9/23	Completed
Questionnaire Form Preparation	20 days	Tue 24/10/23	Sat 18/11/23	Completed
Ethical Form	2 days	Sat 18/11/23	Mon 20/11/23	Completed
Questionnaire Form Collection	3 days	Mon 20/11/23	Wed 22/11/23	Completed
ACKNOWLEDGEMENT	1 day	Thu 16/11/23	Thu 16/11/23	Completed
ABSTRACT	1 day	Thu 16/11/23	Thu 16/11/23	Completed
CHAPTER 1: INTRODUCTION	**2 days**	**Fri 17/11/23**	**Sun 19/11/23**	**Completed**
Introduction	2 days	Fri 17/11/23	Sun 19/11/23	Completed
Problem Background	2 days	Fri 17/11/23	Sun 19/11/23	Completed
Project Aim	2 days	Fri 17/11/23	Sun 19/11/23	Completed
Objectives	2 days	Fri 17/11/23	Sun 19/11/23	Completed
Scope	2 days	Fri 17/11/23	Sun 19/11/23	Completed
Potential Benefit	2 days	Fri 17/11/23	Sun 19/11/23	Completed
Overview of the IR	2 days	Fri 17/11/23	Sun 19/11/23	Completed
Project Plan	2 days	Fri 17/11/23	Sun 19/11/23	Completed
CHAPTER 2 LITERATURE REVIEW	**2 days**	**Mon 20/11/23**	**Tue 21/11/23**	**Completed**
Introduction	2 days	Mon 20/11/23	Tue 21/11/23	Completed
Domain Research	2 days	Mon 20/11/23	Tue 21/11/23	Completed
Similar Systems/ Works	2 days	Mon 20/11/23	Tue 21/11/23	Completed
Technical Research	2 days	Mon 20/11/23	Tue 21/11/23	Completed
Summary	2 days	Mon 20/11/23	Tue 21/11/23	Completed
CHAPTER 3 METHODOLOGY	**2 days**	**Wed 22/11/23**	**Thu 23/11/23**	**Completed**
Introduction	2 days	Wed 22/11/23	Thu 23/11/23	Completed
System Development Methodology	2 days	Wed 22/11/23	Thu 23/11/23	Completed
Data Gathering Design	2 days	Wed 22/11/23	Thu 23/11/23	Completed
Analysis	2 days	Wed 22/11/23	Thu 23/11/23	Completed
Summary	2 days	Wed 22/11/23	Thu 23/11/23	Completed
CHAPTER 4: CONCLUSION	**1 day**	**Fri 24/11/23**	**Fri 24/11/23**	**Completed**
Achieved, justification of research quantity	1 day	Fri 24/11/23	Fri 24/11/23	Completed
Gaps in research and design	1 day	Fri 24/11/23	Fri 24/11/23	Completed
Appendix	**2 days**	**Fri 24/11/23**	**Sat 25/11/23**	**Completed**
Project Proposal Form	2 days	Fri 24/11/23	Sat 25/11/23	Completed
Ethics Forms	2 days	Fri 24/11/23	Sat 25/11/23	Completed
Log sheets	2 days	Fri 24/11/23	Sat 25/11/23	Completed
Gantt Chart	2 days	Fri 24/11/23	Sat 25/11/23	Completed
Respondence Demographic Profile	2 days	Fri 24/11/23	Sat 25/11/23	Completed

Table 1 shows the project planning.

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

To further evaluate the topic “Automated Course Seeking AI-Chatbot using Natural Language Processing (NLP)”, the researcher will explore the implementation of chatbots with natural language processing from the first chatbot to the modern usage of chatbots in society. The researcher will start with a brief introduction of the domain AI, chatbots and natural language processing, then will discuss the development status and the latest implementation in the industry. The usage of natural language processing will be explored as well to have a brief idea of the usage of natural language processing and how chatbots and natural language integrate and trained using machine learning.

2.2 Domain Research

Online learning platform

There are many online learning platforms available such as Coursera and Udemy are known worldwide. Udemy alone has at least 213,000 online courses including professional certifications and degrees in different languages available for users worldwide to purchase from. (Hurley, 2023) During the pandemic times on 2021, the number of online courses increased by 60% compared to 2019, and over 6000 courses were added each month. (Young, 2021) This will cause a problem on course seeking to get the best course for the learners. Other than that, certain fields such as Information Technology develop and eliminate the technology in a very short time. Technology proposed by big companies such as .NET from Microsoft, will have major updates on their .NET development release every year and the knowledge will be outdated and need to add on the knowledge new features. (Gewarren, 2023)

Course Recommender System

Academic advising systems (AAS) are specialized in helping students to get the best personalization and informed decision-making for their future careers. According to research by Algarni and Sheldon (2023), they have analyzed several research on finding the best approach of a recommender system to give the best course selections for the students. These approaches are collaborative filtering, content-based filtering, hybrid recommender systems, and unique (novel) approaches are all efficient ways to create course recommendation engines for the platforms. Some of the methods require prior Academic results to make recommendations. Nevertheless, all of the recommenders need to get prior information about the students before they receive a course recommendation result. Here is where the filter features come from on the websites.

Artificial Intelligence (AI)

At the start of this century, the concept of artificial intelligence has been implemented in every aspect of our lives. From computer processing chips on self-driving cars to virtual assistants like Siri. According to Copeland (2023), artificial intelligence is the ability of a digital computer to perform tasks commonly associated with intelligent beings. In simpler term, AI is enabling computers to do smart things where AI have multiple brunches of technology to develop. In the year 2023 alone, AI tools such as Voice and speech recognition, AL-powered cybersecurity and generative AI which can generate video and pictures from text are developed based on the one to two principle of the AI branch. (Shenwai, 2023)

Natural Language Processing (NLP)

Natural language processing is a branch of Artificial Intelligence to allows computers to read and comprehend information from human sentences. NLP consists of two main subsets: Natural language understanding and natural language generating both of them have multiple complex processes and analyses to extract meaning, predict the needs of the user and generate a response upon the prompt. (Ai, 2022)

Implementation of Natural Language Processing (NLP)

These powerful chatbots have a primary ability to have before performing the tasks, that is to comprehend the needs of the human input and give the best right answer. The NLP is the core technology that is used to tell the computer what to do with human sentences. NLP is being used in other fields such as user posting on social media platforms. According to research by Valdez et al. (2023b), they have used NLP to analyze tweet postings on X, formerly named Tweeter, about vaccines for misinformation indicators during the COVID-19 times. The NLP analyses support two different languages English and Spanish. In conclusion, they found misinformation in both languages, and some of them overlap and they have provided misinformation in different areas related to vaccination distribution. This is one of the implementations of NLP used for analysing human posting sentences.

Machine Learning (ML)

Machine learning is a branch of AI and computer science which focus on the use of data and algorithm to imitate the way that human learns by the result improving the accuracy of completing a certain task assigned. (IBM, n.d.) A good example of the implementation of ML is the use of statistical methods and algorithms that are trained in the model used in the data science field, the computer is enable to make classification or predictions and to uncover insights in data mining projects. They help in making business decision making with the provided insight.

Machine Learning Algorithm

There are three types of machine learning algorithms: supervised learning, unsupervised learning, and reinforcement learning. Supervised learning uses a labelled dataset to train with predefined categories that make the machine learn a mapping from the input data to the output label, so it can make predictions. (GeeksforGeeks, 2023) Common frameworks used to create algorithms would accelerate solution development such as TensorFlow and PyTorch. Training models for each specific usage of chatbots requires different types of training methods, as for this project, the researchers will use the supervised method to complete the chatbot.

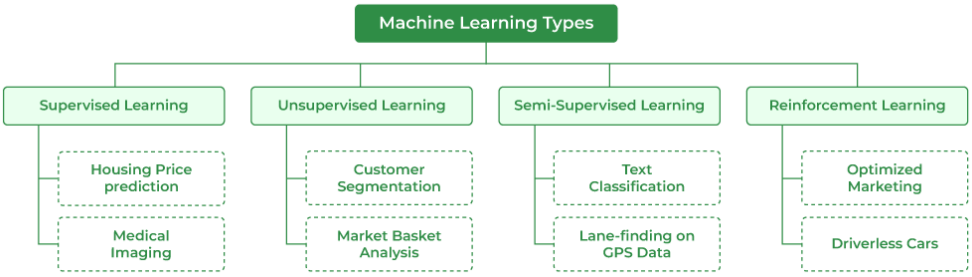


Figure 1: Types of machine learning and sample on implementations. (Machine-Learning-Types, n.d.)

Quantum Machine Learning

Natural language processing has data processing phrases certain techniques are used after tokenizing the sentences for the computer to process. The traditional ways of processing tokenized words are by logistic regression. (DeepLearning.AI, 2023) However, Chatbots such as Google's Barn and ChatGPT have huge NLP models that are commonly called large language models (LLM) to generate texts that are accurate for commercial application use. These LLMs require high computational power and memory that will learn to infrastructure cost. (Barla, 2023) Quantum machines are proposed for use in NLP applications as quantum machines provide the capability of solving problems that normal computers are difficult to solve by using quantum mechanics. Research by Pandey et al. (2023) combined quantum machines and machine learning techniques to compare the performance of using classical computing with ML for parts of speech tagging of code-mixed datasets. They found out that applying quantum to long short-term memory will enhance the performance of POS tagging of social media code-mixed language on multiple platforms. Those algorithms used in quantum machine learning are beneficial to natural language processing applications.

Conversational Chatbots

The first ever chatbot named Eliza was first introduced by Weizenbaum (1966) four decades ago. Eliza is a chatbot from a computer that can give a response upon user inputs that's first even made so computers and humans can have a conversation together. The concepts of the reply mechanism focused on how a computer can understand the prompt entered by the users, which are identification of keywords, discovery of minimal contexts, choices of appropriate transformations, generation of responses in the absence of keywords and finally provision of an editing capability for ELIZA's replying to scripts. These concepts formed the fundamental procedures of machine attempts to digest natural language, and they are being used in the natural language processing (NLP) development concepts.

Chatbots integrate with natural language processing

The researchers have shown the application of NLP for different uses, however, chatbots are machines that use NLP together with artificial intelligence that can improve accuracy and do much more. According to research by Dongbo et al. (2023), they pointed out that many pieces of research related to chatbots have one issue, that is those chatbots do not accurately identify the user's requirements. Hence, they have proposed a chatbots using Bi-directional recurrent Neural Network with a Fuzzy Naïve Bayes classifier(BRNN-FNB) that helps in identifying the requirements and an AI-based sentimental analysis to perform accurate interactions. The result of the research is they got a 93% accuracy of the proposed chatbots with BRNN-FNB using the seq-to-seq techniques, and 92% per cent of accuracy without seq-to-seq techniques. The seq-to-seq model is a class of Recurrent Neural Network architecture that is used to solve complex language problems such as creating chatbots.

Implementation of AI-Chatbot as an interactive avatar in a school

Nowadays, chatbots are greatly advanced and they are being used in many fields, among the chatbots conversational chatbots are the most popular ones. These chatbots are equipped with Natural language processing to allow the computer to comprehend human languages and use deep learning techniques to allow responses with decent answers. For instance, a Xatkit development framework-based chatbot system architecture proposed by Baha et al. (2022) is capable of receiving string and audio inputs, processed by an in-house

NLP engine to process intent detection and recognition. The chatbot's output is not a computer-generated plain text, but a 3D avatar can react with certain gestures, and mimics and continue the conversation with the audio playback of the generated responses. This chatbot is integrated with multiple modern technologies of NLP, deep learning, intent recognition, and a Speech Synthesizer module to turn text into speeches that could sync with the gesture of the avatar. This Edu chatbot is implemented in the classrooms of schools.

Implementation of chatbot applications and their limitations in the education field

A systematic review of chatbot applications in the education field was carried out by Okonkwo and Ade-Ibijola (2021), they found many articles focused on chatbots are useful as a learning aid in teaching and learning such as personal learning, automation of administration work, aid in assessment, advising on educational issues, research and development aids. For the limitations, they found that chatbots have several issues with ethical problems of artificial, accurate evaluations, user attitude, chatbot supervision and difficulty in maintenance. In the end, the researchers proposed future research work for the research domain to further enhance the chatbots with technological advancements, completing the ethical principles developments of artificial tools so they will be safe to use in daily life as well as usability testing. As chatbots are further enhanced and grow, it is necessary to have well-defined rules for using chatbots at work so they will be compatible with user ethics.

Limitation migration of AI chatbots in education field

According to research by Steele (2023), she has pointed out three major threats of ChatGPT to the students in education performance and added her counterpoints to those issues. The issues are the field of measurement of the student's actual ability, providing information inaccuracy and skill devaluation because the GPT can do a lot of writing for humans in just a few seconds. On the other hand, she emphasized harnessing the potential of these tools that can help the students to learn, as well as mitigate the threats, as the AI could also foster the critical thinking of the students when they use the ChatGPT to comprehend something. They give up rapidly. In the end, this researcher thinks that we must make use of these tools to forge the future.

AI-chatbot in business information system

Chatbots are useful to implement in business intelligence systems(BIS) as well. According to Azmi et al. (2023), they have done case studies on AI with BISs where AI brings many powerful and convenient capabilities to the systems. The capabilities such as enhancing data analysis and providing predictive analysis as well, as automating data integrations by providing natural language interfaces, detecting anomalies in the operations, enhancing data visualization into charts and diagrams and delivering personalized recommendations for users. They proposed a flowchart on the business information system, and where and what AI capabilities can be used to ease the operation of BIS and increase the accuracy as well.

Conclusion

The technology, concepts and implementation of chatbots and natural language processing from the first chatbot in 1966 to the modern advanced chatbots are discussed. Some issues and solutions of AI are pointed out. Some solutions are being implemented to fix the issues such as quantum machine learning to fix memory and computing power lacking problems. Examples of AI chatbot implementations in the educational field are being discussed. In conclusion, AI chatbot with NLP has great potential to enhance the quality of work for human society. Although technical and ethical issues arise when one breakthrough is made, the chatbots are nevertheless helped in human society in many ways. In this project, we will implement the AI chatbot to help in solving SDG 8, to provide sustainable work opportunities for everyone by allowing them to have the best learning opportunities with the help of technology.

2.2.1 Similar Systems

Similar Conversational Agent (Chatbots)

Table 2: Similar works in Conversational AI-Chatbot used in advisory.

Authors	Domain	Model Training Methods	Technic and data model used	Key Features	Findings	Difference with Seeking Chatbot.
Baha et al. (2022)	Education	Use prebuilt model from Xatkit.	Intent Recognition module using CamemBERT encoder and intent recognition, Google Speech Synthesizer API, and 3D Avatar Rendering Engine.	Student supporting chatbot with speech and avatar.	Shown prove that multifunctional media system can work with AI chatbot.	The chatbot uses Google Speech Synthesizer but no Avatar engine. BERT language model is used instead of CamemBERT.
Choudhary and Chauhan (2023)	General	-	Bahdanau attention mechanism, BOW model for vectorization. Beam search decoding for response prediction. Neural network with long short term memories and Seq2Seq architecture.	Increase the accuracy of the chatbots' response.	Bleu-4 score of 0.8537285. High accuracy but slower response time.	It uses RNNs attention mechanism, where the chatbot uses Bidirectional RNNs to process the informations.

Dwyer et al. (2023)	HealthCare	Supervised Machine Learning with labelled dataset involving combinations of intents and keywords.	Keyword intent Recognition.	Provide automated guidance, ask follow-up questions and forward the message to the care team.	Accuracy of chatbot is inadequate. Further testing are needed before deploying.	Used Bidirectional RNNs to fine tune the chatbot increase the accuracy of responses. Will navigate through pages in the system.
Birkun and Gautam (2024)	HealthCare	Using pre-built Large Language Model API.	ChatGPT	Provide instant health aid guide during an heart attack.	Will provide misinformation to user.	Not using LLM API but will train with whole model using BERT.

From Table 2, the researchers found out the focus of performance on the chatbots are accuracy and responding time. If the chatbot is required to be used in mission-critical or life-critical domains such as health-aid during heart attack research by Dwyer et al. (2023), the focus of the users will be accuracy. Other than that, for interactive chatbots such as being used in education as a learning helper, the response time is equally important as the accuracy to give a decent experience for the students. Many different techniques and methods are used to increase the accuracy of responses, and the best accuracy chatbot uses neural networks to train the models to achieve the best accuracy.

Low accuracy of LLM does not conclude LLM are not powerful, but they need a knowledge graph specified to a certain domain through fine-tuning. The powerful part of the LLM is natural language understanding. By adding up-to-date knowledge and giving the complete incorporation of knowledge graphs, they are already tailored to be specifically accurate in a certain domain. However, proper testing and measurements are compulsory before fully deploying AI to handle such things.

In summary as stated above in difference column, more detailed on the algorithm and method on models and intent recognition techniques are identified.

2.3 Technical Research

2.3.1 Physical Hardware

In this section, the researcher will identify the physical requirements for developing this project and the software and tools required for training the chatbot with justification.

There are some components needed to build the system and train the chatbot. A coding environment with the selected programming language and a platform to train the chatbot. Despite on choosing any of the tools, the minimum requirement for the hard disk will be 512GB Solid Secure Disk (SSD). It is more than enough to create a project with the IDE installed and fit the trained chatbot into the computer. In terms of training the neural network of the model, it consumes a lot of RAM and GPU RAM. Computer RAM is to store the whole running project, and the GPU RAM determines the speed of training the model. However, due to the small dataset being used, the integrated graphic card would be enough to train, and RAM will be a minimum of 16GB will be adequate. Next will be a processor, a 6 to 8-core processor with a minimum base clock speed of 2.3GHz will be enough to run the programs.

This is the summary of the specification the researcher will be using:

Processor	Ryzen 5000 Series 6 Cores @ 2.30 GHz
RAM	16GB DDR4
Storage	512 GB SSD
GPU	AMD Radeon ™ Graphics

2.3.2 Programming language

The most common programming languages used to develop chatbots are Python and Java. At the table below will show the comparison: (Staff, 2023; GeeksforGeeks, 2023a; Miller, 2022)

Table 3: Comparison for Python and Java Programming Languages.

Characteristic	Python	Java
Code	Generally fewer line of code, simple and flexible	Generally longer line of code and straight forward. Limited control on garbage memory allocation.
Syntax	Compared to be more like human language.	Syntax rules are strict and use of exception to detect error is required.
Framework	Compared to java, lower number of framework.	Large number of frameworks.
Available of Libraries	Many available related to machine learning and natural language toolkit.	Many third-party libraries for ML and NLP available.
Deployment and Portability	More commonly deployed desktop and web application. Less usable in mobile devices.	Can run on any device or embedded system with Java virtual machine (JVM).
Performance	Slower	Faster due to supporting multithreading.
Object oriented	Yes	Yes
Key Features	Less line no of code, Rapid deployment, and dynamic typing.	Self-memory management, Robust and Platform independent

Result of the comparison, Python has the advantage of the library support to machine learning and natural language toolkit as well as rapid and flexible deployment that is needed for the project. Python is excellent for building machine-learning apps. (Hartman,2023) Hence, Python will be the programming language used for developing the system.

2.3.3 Libraries and tools

Datasets and user input have to be pre-processed before being to the training where is the machine learning phrases. The two libraries of Python used for text processing are natural language toolkit (NLTK) and Machine learning using scikit-learn.

Natural Language Toolkit (NLTK) Library is widely used to develop NLP applications and analyse text data. Its interface can carry out the basic tasks for language processing such as tokenization, stemming, lemmatization, parsing, and sentiment analysis. The NLTK can

effectively manipulate and analyze linguistic data. (Python, 2022) It also can use the collection of corpora that provide a rich data source for training and testing the NLP models. (Ali, 2023)

Scikit-learn is a Python module that provides machine learning algorithms for medium-scale problems, both supervised and unsupervised. This module brings focuses on ease of use, performance, documentation, and API consistency. It allows machine learning for non-specialists using Python. (Pedregosa et al., 2011)

2.3.4 Testing tools

Botium is the chatbot testing tool the research will be using. Botium is an open-source tool it has a large community to further develop using Botium on GitHub. It is used to ensure the chatbot is working as expected. (*What Is Botium — Botium Documentation*, 2023)

2.3.5 Operating System

The common operating systems are Windows, MacOS and Linux. There are several key elements to deciding which chatbot platform compatibility with OS, programming language compatibility, program hosting requirement, cost and developer preference. As the chatbot is planned to run on small computers, a server operating system is not required. In terms of costing, if there is no operating system being purchased, a free Linux OS is the best choice.

The operating system that will be used is Windows 11 by Microsoft. Windows 11 has many benefits including a clean and user-friendly interface and supports almost all of the file formats on the computer. It also brings copilot AI into the device to help users get better search results on their computer. (Microsoft, n.d.) It is also compatible with Visual Studio with Python, as well as all other tools stated in the software requirements section.

2.4 Summary

In summary, physical and software requirements are discussed and justified in this section. These tools are available on the internet and it able to be used in this project. All of the minimum requirements are listed and met by selecting the right tools and infrastructure for this project.

CHAPTER 3: METHODOLOGY

3.1 Introduction

The methodology in a project plays a vital part in progressing through the whole project. It decides how to carry out the tasks in each project section. The whole project will be a methodology. The methodology will be carefully selected to fit different use cases. In this section, the researcher will explain system development methodologies, the methodologies available and the justification of selection in this project.

3.2 System Development Methodology

3.2.1 Introduction of methodology

Methodology refers to the methods and techniques used to reach a certain goal in a project or a whole list of tasks, it is referred to as the system development life cycle(SDLC). (Pedamkar, 2023) In this project, the researcher aims to develop an AI-chatbot with an NLP system to help learners in course seeking. Hence, justifying the problem statement and the feasibility of the chatbot in solving the problem is important before diving into the system development. There are several methodologies available for this project. The table below states the comparison of these methodologies: Water, Rational Unified Process (RUP) and Extreme Programming (XP). (This vs. That, n.d.; Zaminkar, 2013; Zaminkar & Reshadinezhad, 2020; Khan et al., 2021; Shrivastava et al., 2021)

Table 4: Comparison between Waterfall, RUP and XP.

Characteristic	Waterfall	RUP	XP
Traditional or Agile	Traditional	Agile	Agile
Development Process	Sequential	Iterative and incremental	Frequent iteration and have multiple releases.
Needs of experts	Not necessary	Requires ensuring quality and complete within the time span.	Required ensure effective collaboration.
Management	Easy to manage because of rigid phrases.	More complicated because the phrases are iterative and disorganised.	Required communication with client to meet the deadlines.
Time span	Longer	Shorter	Shorter
Collaboration	Less collaborative required because the phrases are predictable.	Close collaboration required in each development iteration.	Collaboration is more frequent between developers and the client to

			product small release from time to time.
Characteristic	Waterfall	RUP	XP
Adaptable to change	Less adaptable	More adaptable	More adaptable
Key feature	Each of the stage is divided clearly and is easy to implement by beginner. It is suitable for most of the projects with longer time frame.	Enable adaptability for changing requirements during the development process.	Focus on adaptability of changes and produce small releases frequently. Focus on enhancing coding, communication with the client, create releases that suits the requirements.
Limitation	Less adaptability on change. Consume large resources to adapt in a small change.	Tend to delay or fail the project due to multiple iteration of the project.	Lack of documentation due to emphasizing on the releasing of project more than producing designs of system. Complicates development progress when the team is larger. May result in never ending project without proper time frame management.

3.2.2 Methodology Choice and Justification

From the table above, the RUP is the most suitable methodology for this project. As the project progress, meeting with supervisor and change in system requirement might occurs. Hence, with multiple iteration on developing the system and get feedback will be suitable for the characteristic of this project. With proper management with the progress, the project can be completed within the time frame.

3.2.3 Activities and Processes

RUP methodology is an incremental software development process with four main phases: Inception, elaboration, constructions, and transition. Each of the phase is a sequential and iterative approach of developing a product. RUP is also a mindset with 9 principals. (Paul Dragos, 2021) The researcher will go through the detailed tasks in each of the phrases.

Inception Phase

The first phase is to do research about the domain and construct project aims and objects. Domain research to gather insight into the field of course-seeking platforms and tools, AI, NLP and ML. Integration research of these domains is done to find their relationships and implementation in the industry. The design of the survey is done to collect feedback on existing online learning platforms and the experience of course-seeking tools. The survey is to find the current problem and adjust the proposed system requirement according to the responses. After analysis, the project aim, objective and problem statement are justified. Project schedule planning is the last deliverable for this phase.

Elaboration phase

The framework used for developing the chatbot system is evaluated in this phase including design system architecture. Library, testing tools, modules and datasets used are identified in this phase. Starting with a list of recommended libraries and testing tools, which are listed in a table to do a comparison of the key features, and then the best choice will be justified from the comparing table. The final system designs with the required tools and library are documented. The final system design needs to align with the aim and objective of the project.

Construction Phase

This phase starts with the actual coding. The system is developing with the design created using the planned programming language for the system. The labelled dataset will be inserted and used to train the model. The interface is built using Python, and GitHub is used for version control and progression tracking. The requirement of the system is aligned with the aim and objective of the project.

Transition Phase

After delivering the system source code, unit testing is first carried out to ensure intended responses from the system. User inputs and triggering such as positive and negative values, string values and symbols are used to test the input components of the system. All of

the situations need to be handled properly, especially for invalid inputs. It is to prevent the system from crashing. After that, integration testing is done by combining two components to work together as intended. The final testing will be system testing which will test the system by experiencing it from a user perspective. All the processes in testing are recorded including deflects and successful attempts. If the deflects are too many, the researcher could consider iterating back to the construction phase to debug and systems again with the testing report.

Operation phase

After the system passes the testing, it will be brought to the user site to test run the systems by the targeted users. Quality Assurance testing (QA) will be carried out for deploying the system to the user and examining the feedback from the users. Documentation closing for the project will be done in this phase. Including the System Manual, FAQs and guidelines for the user. This will be the end of the project.

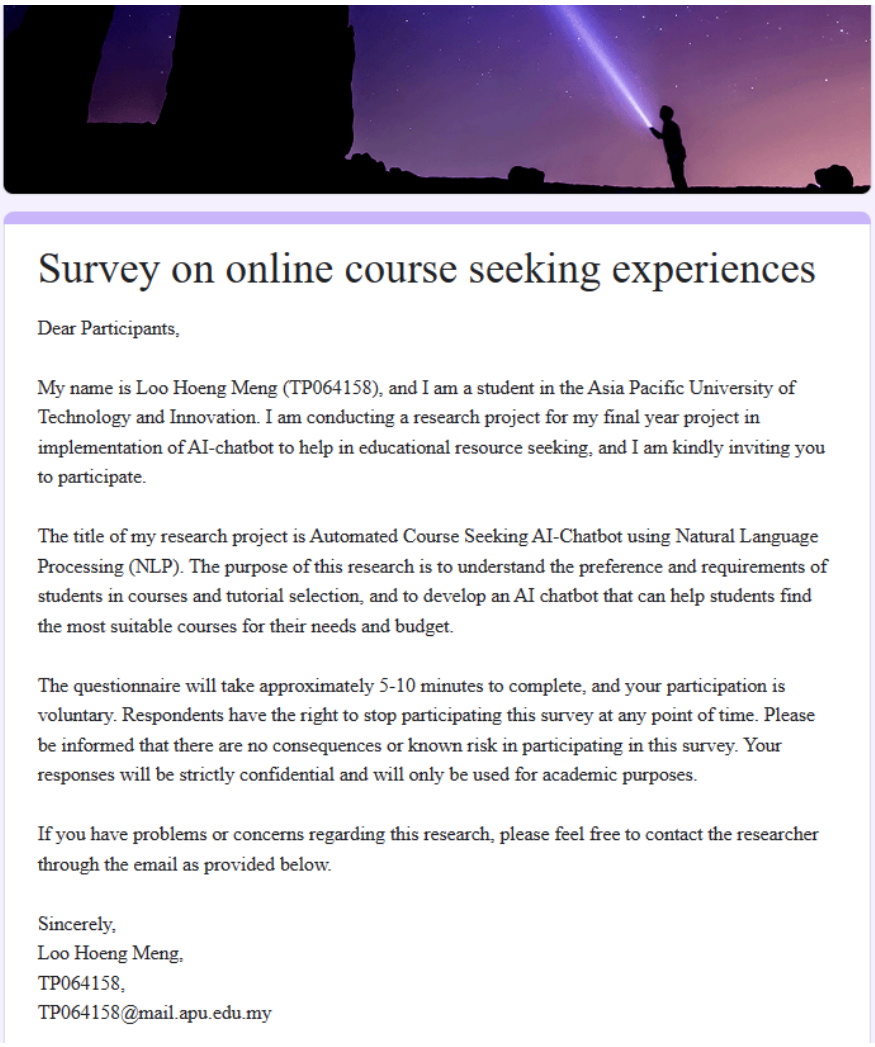
3.3 Data Gathering Design

A survey is a research data collection method that has a list of questions or type of prompts that aims to collect information from a target group or respondents. (Bhat, 2023) The type of questions usually is a mix of open-ended and close-ended questions. The researcher aims to collect qualitative information related to the experience of respondents on getting recommended course-seeking experience on those websites. Hence, the questions are properly designed to get the intended data from 41 respondents, by following the guidelines of the ethic form and approval from the supervisor.

The survey consists of 4 sections with 18 multi-selection questions and short answer questions. The three sections are Introduction, Demographics, Course-seeking Preferences and Requirements, Personal Learning Experiences.

The design of the survey cover page and the 18 questions are displayed below.

Cover Page and Introduction



Survey on online course seeking experiences

Dear Participants,

My name is Loo Hoeng Meng (TP064158), and I am a student in the Asia Pacific University of Technology and Innovation. I am conducting a research project for my final year project in implementation of AI-chatbot to help in educational resource seeking, and I am kindly inviting you to participate.

The title of my research project is Automated Course Seeking AI-Chatbot using Natural Language Processing (NLP). The purpose of this research is to understand the preference and requirements of students in courses and tutorial selection, and to develop an AI chatbot that can help students find the most suitable courses for their needs and budget.

The questionnaire will take approximately 5-10 minutes to complete, and your participation is voluntary. Respondents have the right to stop participating this survey at any point of time. Please be informed that there are no consequences or known risk in participating in this survey. Your responses will be strictly confidential and will only be used for academic purposes.

If you have problems or concerns regarding this research, please feel free to contact the researcher through the email as provided below.

Sincerely,
Loo Hoeng Meng,
TP064158,
TP064158@mail.apu.edu.my

Figure 2: Introduction of survey.

The introduction of the survey has informed the terms and conditions of the research and their rights to participate in the survey. Respondents are informed that they are participating voluntarily, they can stop participating at any point in time, there is no known risk participant in this survey and all the responses are strictly confidential and only used for academic purposes in this project.

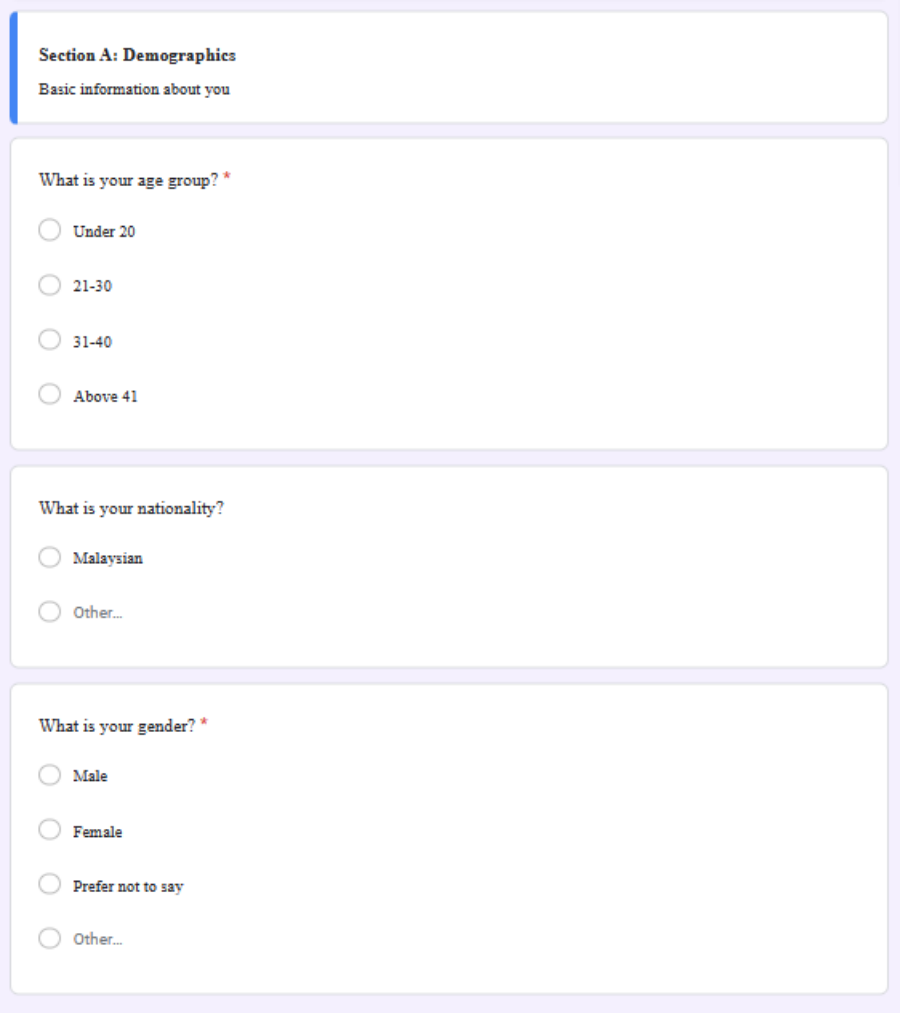


Figure 3: Demographics section.

The demographics will have some brief information about the respondents to aid in the analysis process.

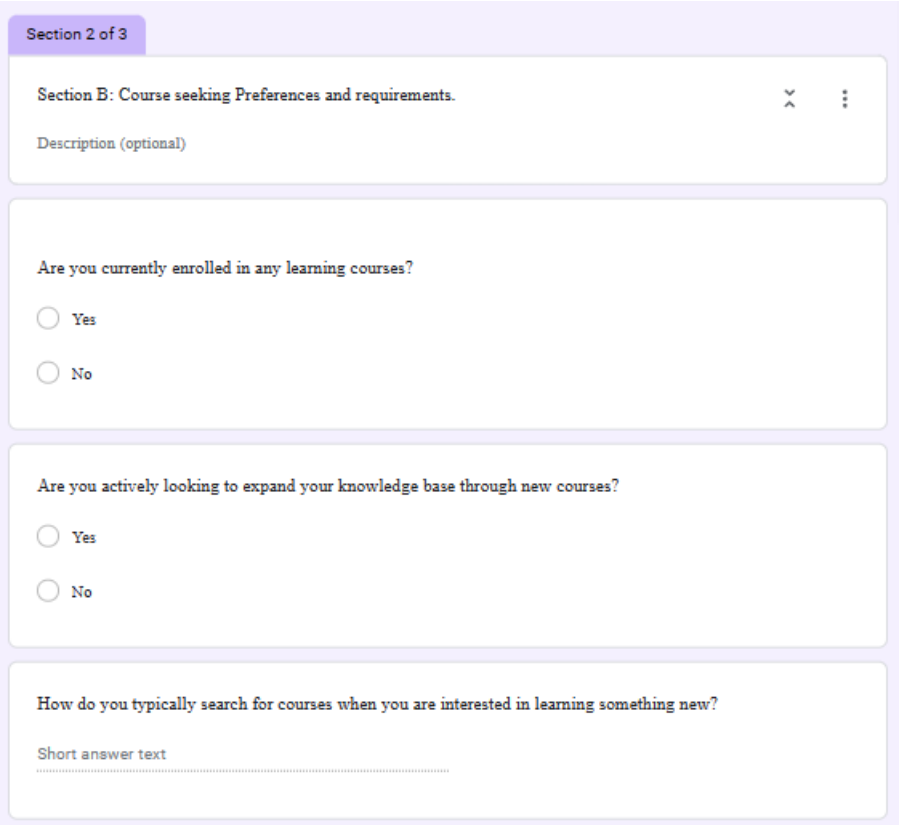


Figure 4: Course seeking preference and requirements. Question 4 to 6.

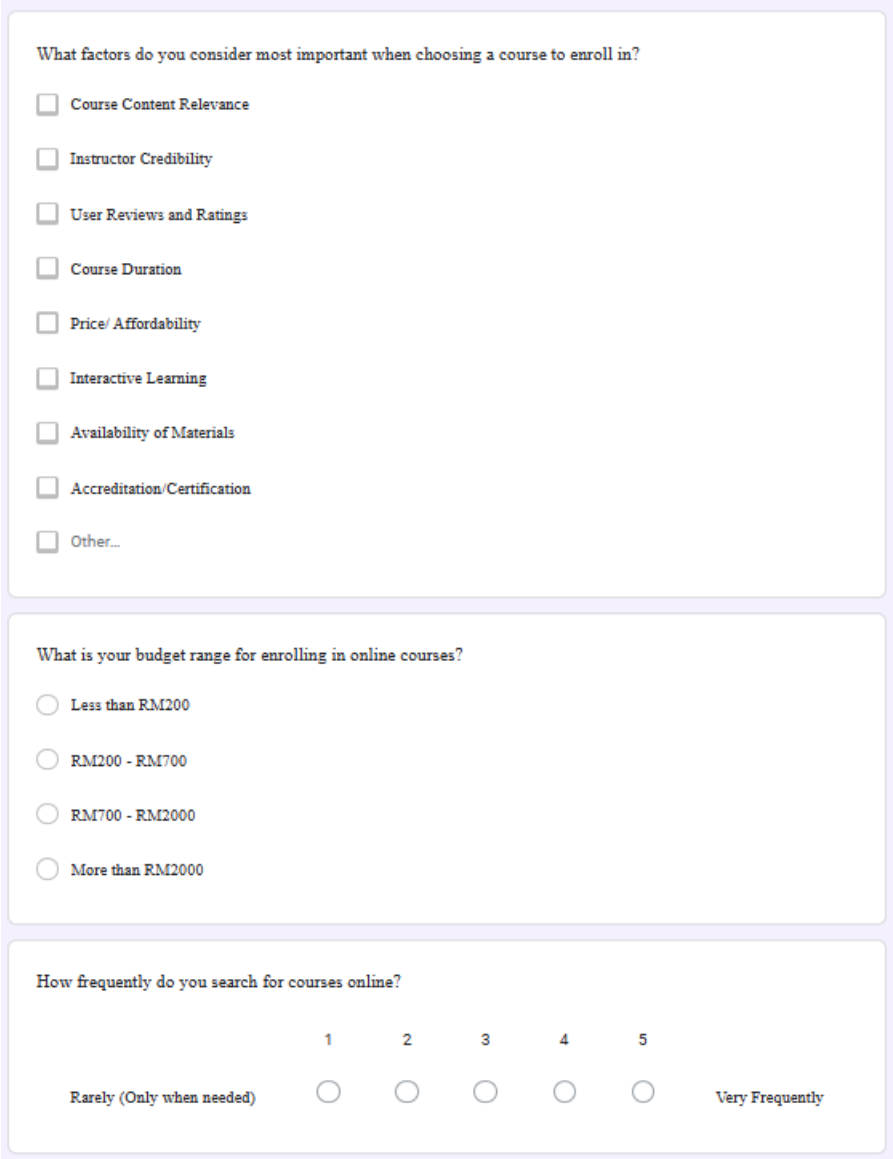


Figure 5: Course seeking preference and requirements. Question 7 to 9.

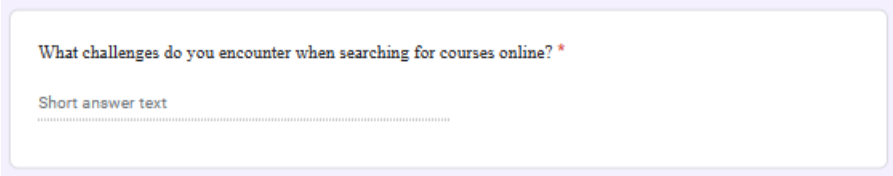


Figure 6: Course seeking preference and requirements. Question 10.

This section focusses on their familiarity with getting courses online and their preferred characteristic of their ideal online course.

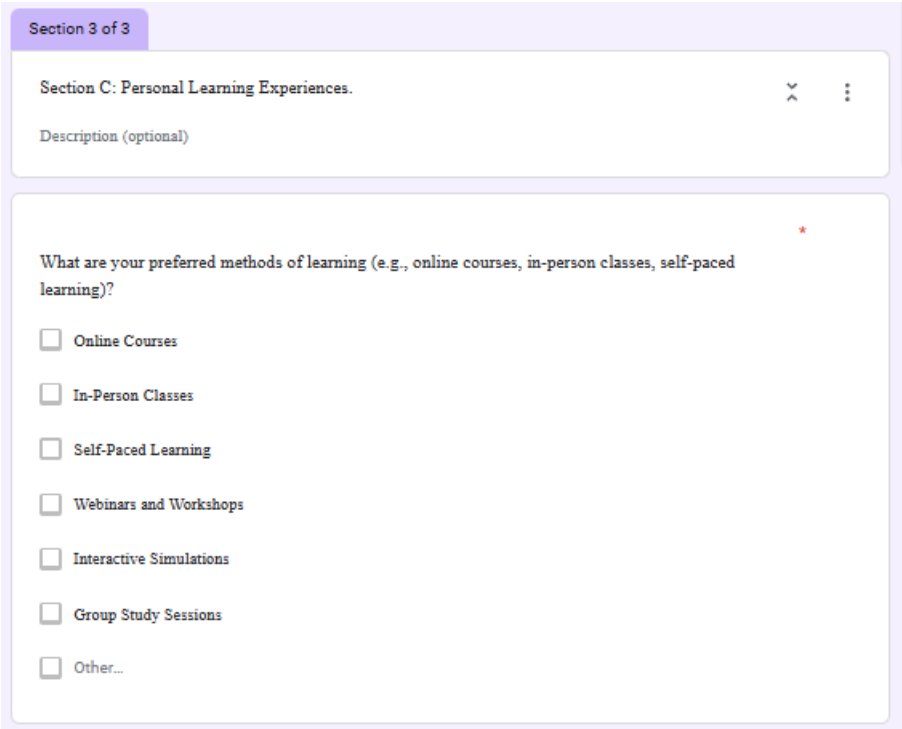


Figure 7: Personal Learning Experience section from question 11.

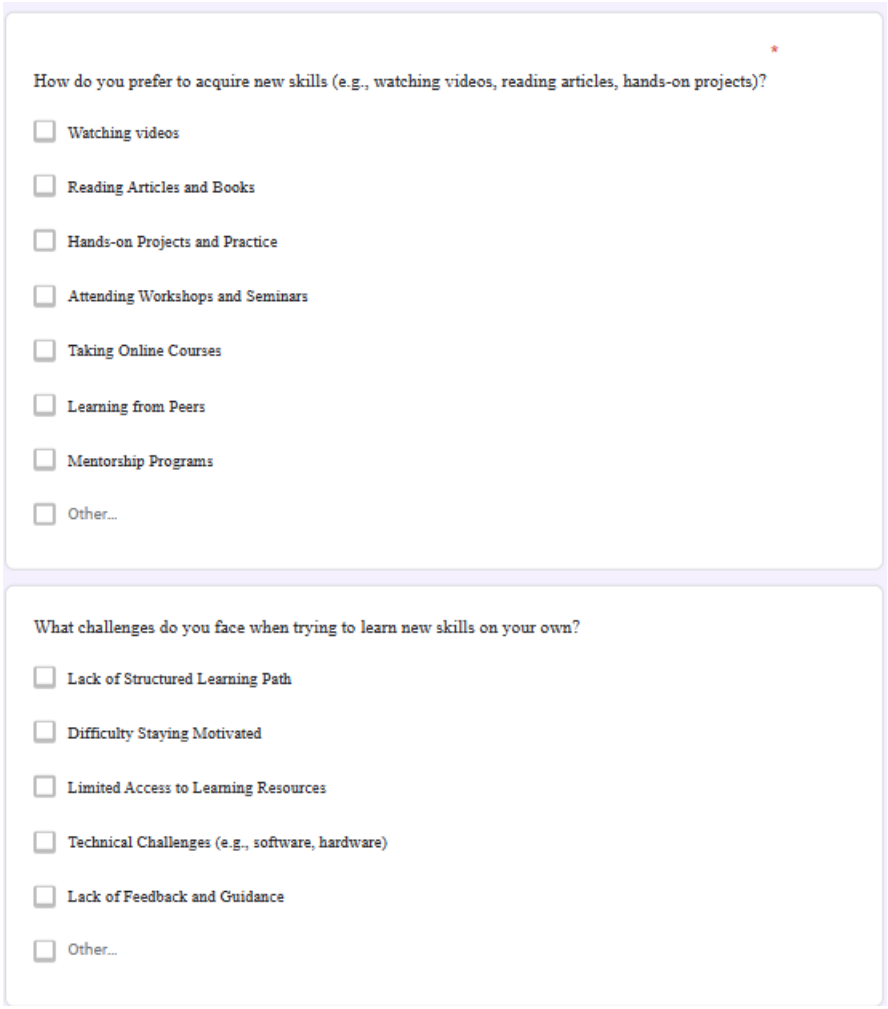


Figure 8: Personal Learning Experience section from question 12 to 13.

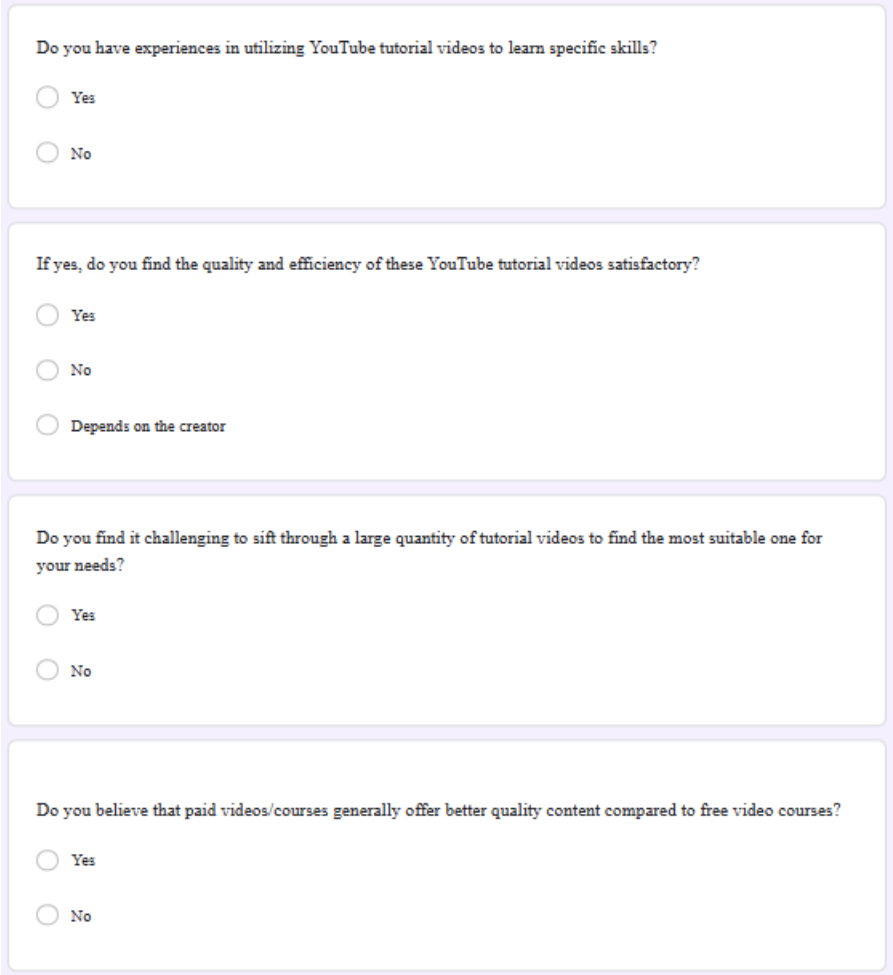


Figure 9: Personal Learning Experience section from question 12 to 17.

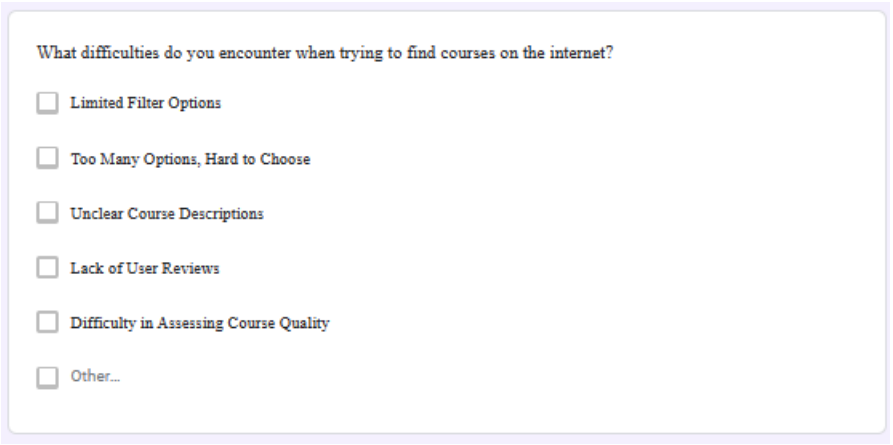


Figure 10: Personal Learning Experience section from question 18.

This section is to find the issue encountered when the respondents trying to get tutorial videos or when they trying to find the best course for them. This is to align the problem statement for this project.

3.4 Data Analysis

The researcher will analyse the 41 responses using the question prepared on the previous section.

Section A: Demographic

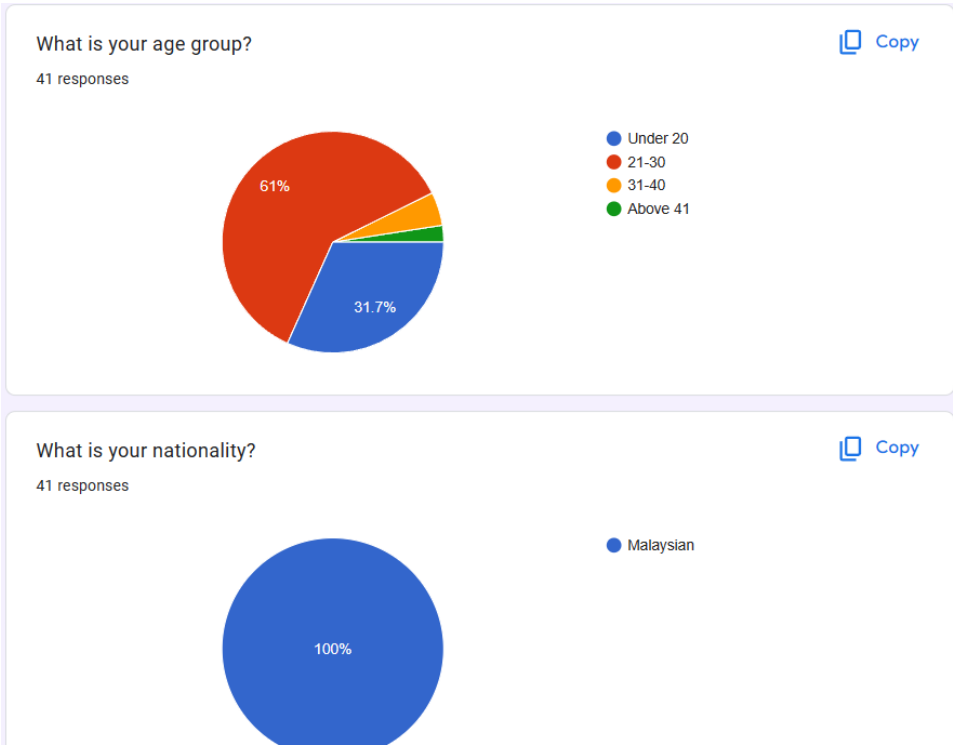


Figure 11: Responses of question 1 and 2.

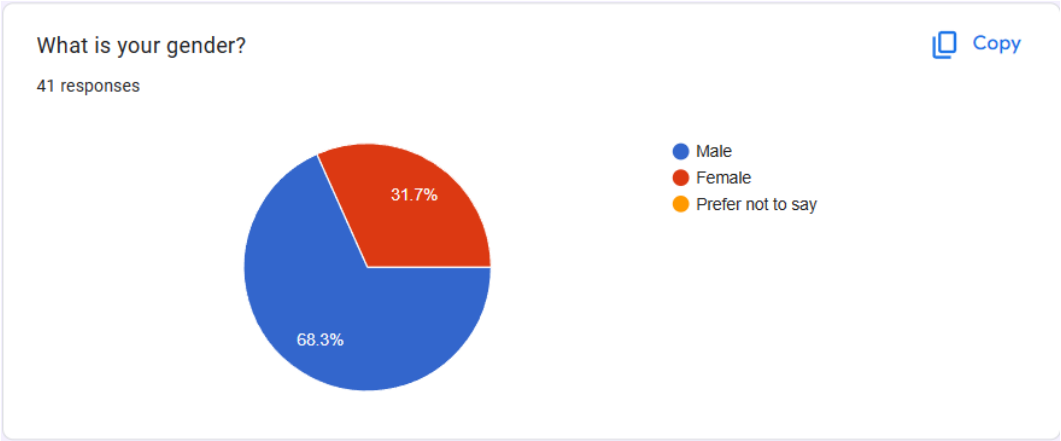


Figure 12: Responses of question 3.

In the demographic analysis of the respondents, half of the respondents are aged range from 21 to 30, and one-third of the respondents are between 18 to 20 years old. Only less than 10% are aged 31 and above. The gender distribution has a ratio of 70 to 30, with a higher representation of males compared to females. All the respondents are Malaysian.

Section B: Course seeking Preferences and requirements.

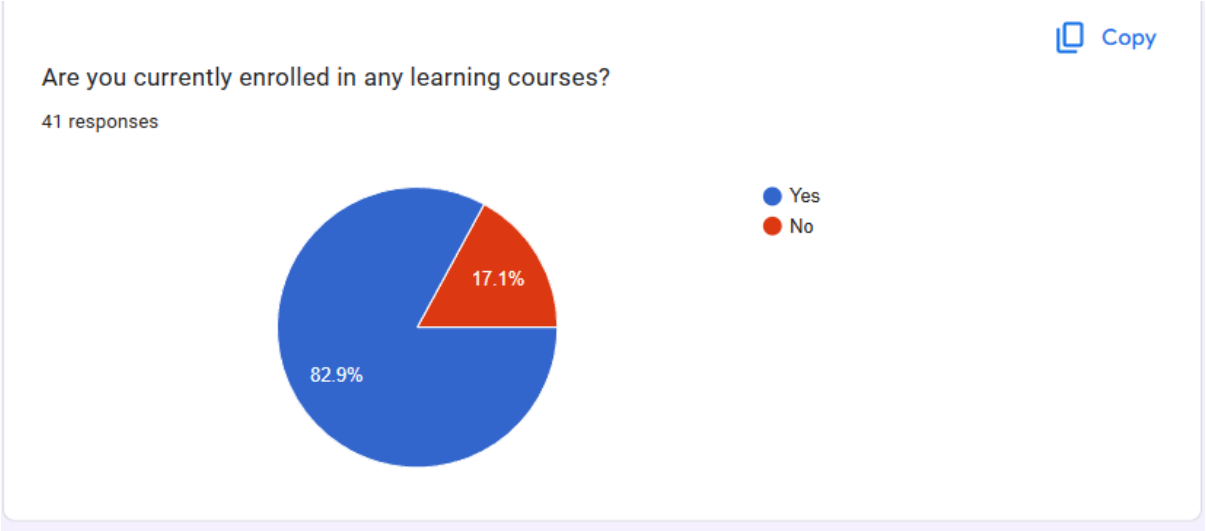


Figure 13: Responses of question 4.

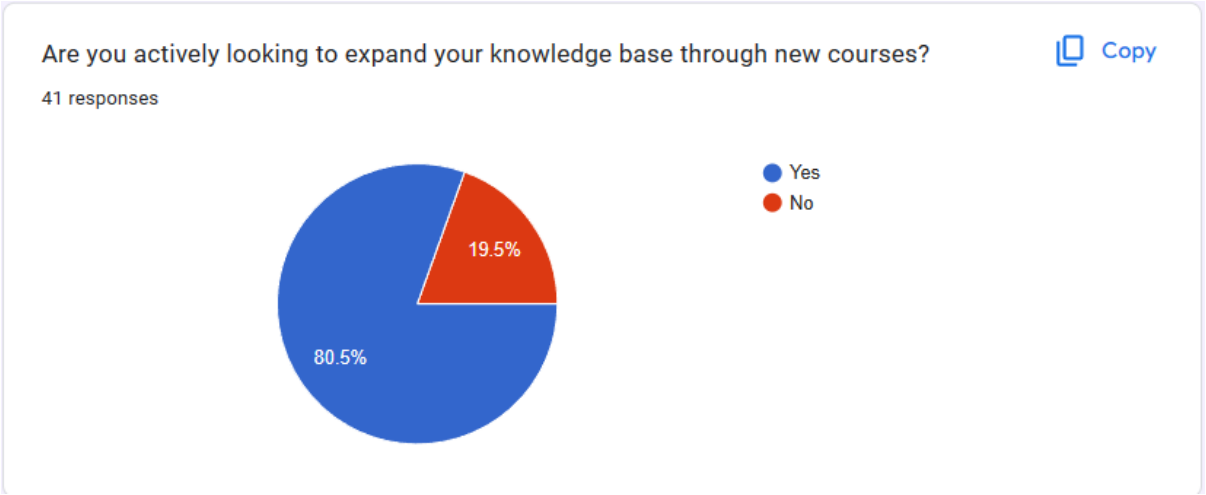


Figure 14:Responses of question 5.

From questions 4 and 5, about 82.9 percent of respondents are learners and 80.5 percent of them are actively looking for new courses to upskill themselves.

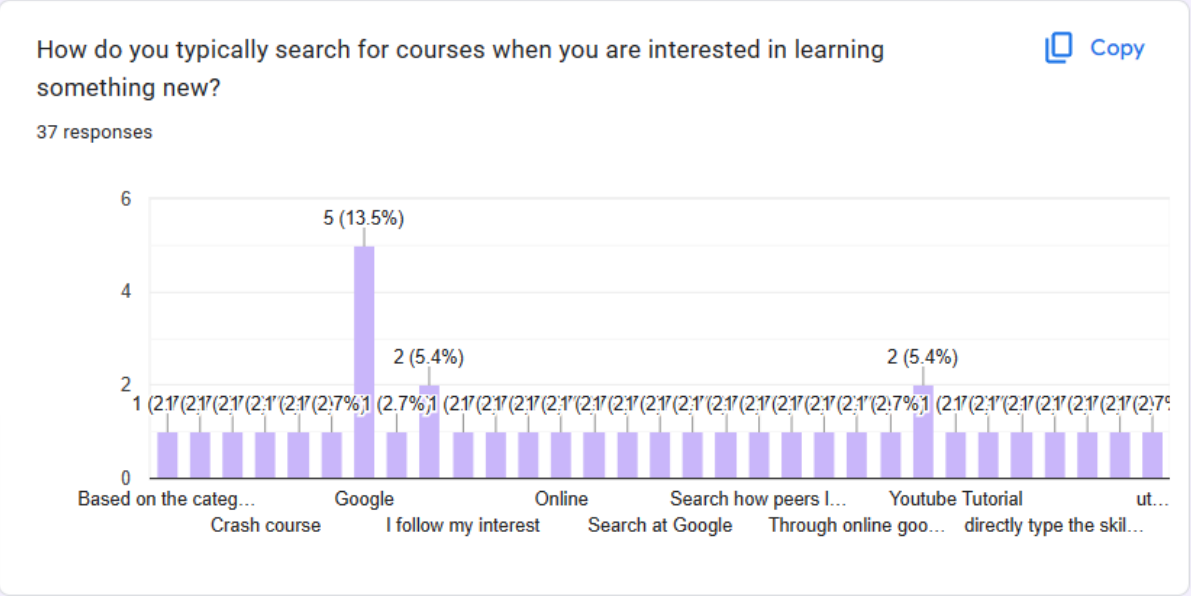


Figure 15:Responses of question 6.

Table 5:Analysis of responses in question 6.

Online / Web / Google / Search Engine	29 Respondents
Through YouTube	8 Respondents
Through Udemy or Coursera	1 Respondents
Broadcasting	1 Respondents
Communicate with senior	1 Respondents
Blank responses, unclear answer	8 Respondents

From question 6, google is the most favourable method to search for online courses with 29 respondents (63%)mentioned about it.8 respondents (19%) used YouTube to check for tutorial videos. About 1 respondents (2%) prefer using Udemy or Coursera directly. Other rare options are to communicate with seniors and through broadcasting in course seeking. They might not have a clearly defined approach in this situation. 8 respondents (20%) left this question blank or had unclear answers.

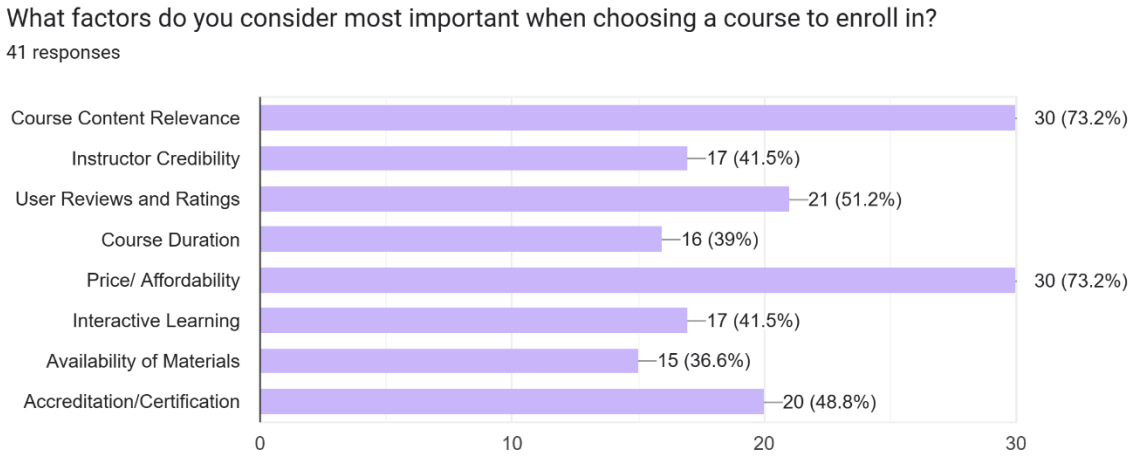


Figure 16:Responses of question 7.

The top three factor the respondents most important while choosing a course is Content relevance voted 30 (73.2%), Price or Affordability voted 30 (73.2%) and User reviews and ratings voted 31 (51.2%). The lease chosen factor is the availability of materials with 15 votes (36.6%).

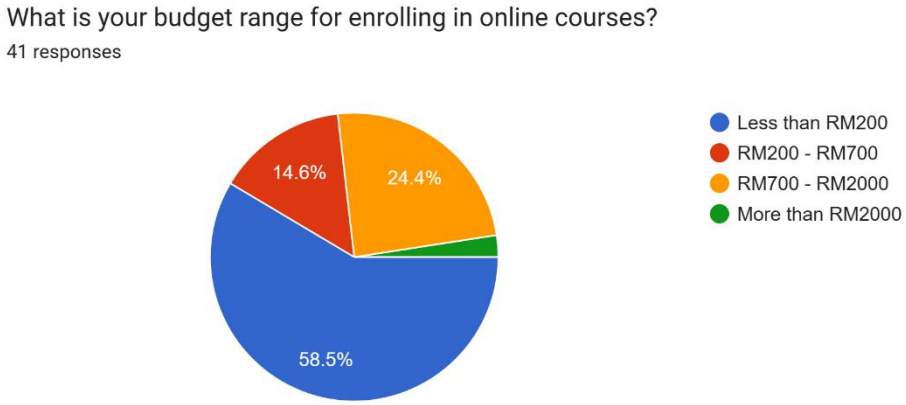


Figure 17: Responses of question 8

58.5% of respondents budgeted less than RM 200 including free courses for online courses, 14.6% budgeted for RM200 to RM 700 and 24.4% budgeted for RM 700 to RM 2000. Only 1.5% of the respondents budgeted over RM2000 for an online course.

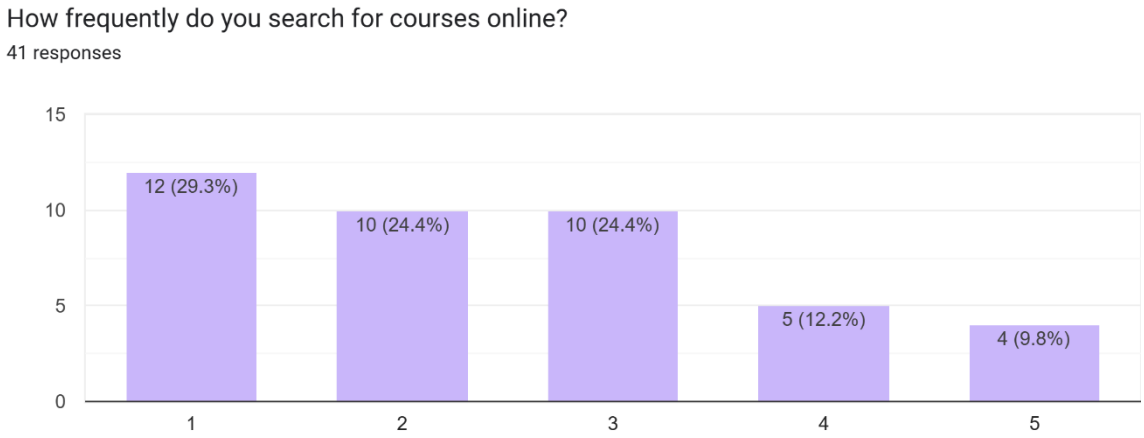


Figure 18: Responses of question 9

Only 9.8% of respondents very often search for online courses whereas 29.3% of people are not frequently searching for courses. Most of the respondents are likely not to search frequently on the internet for online courses.

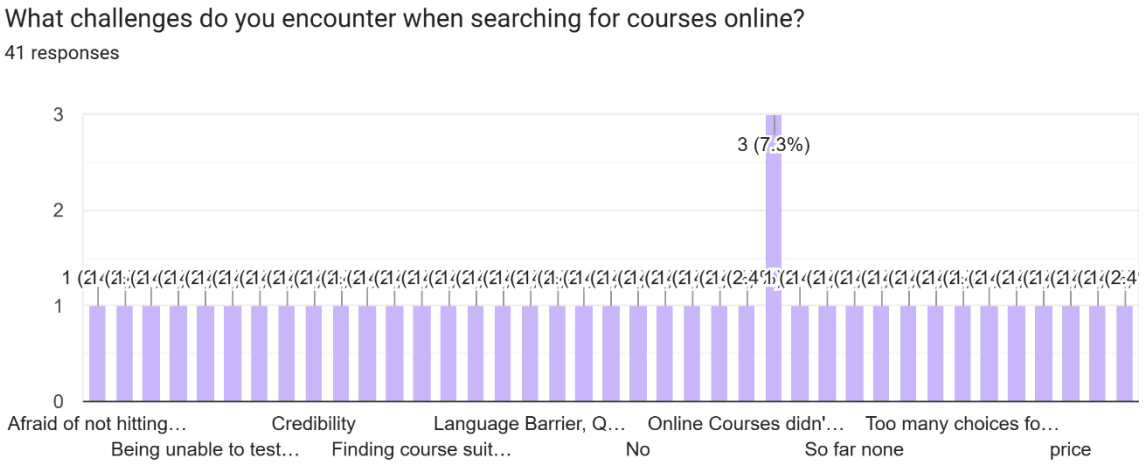


Figure 19: Responses of question 10

From responses to question 10, 13 respondents (31.7%) have faced challenges with the pricing of the course, either due to excessively high costs or an inability to preview the course before paying. Another 13 respondents (31.7%) stated that the abundance of options for a single topic causes difficulty in determining the relevance, reliability, and currency of the content. One (2.44%) stated that the course might not suit the learner's learning style and personality. Two respondents (4.88%) encountered spam on the course learning platform. Additionally, 2 respondents (4.88%) stated that they are afraid the course may not meet the learner's expectations or that they might lose interest in the course halfway. The inability to preview the course was mentioned by 2 respondents (4.88%) who expressed reluctance to commit to payment without previewing the course or having a test run session.

Section C: Personal Learning Experiences.

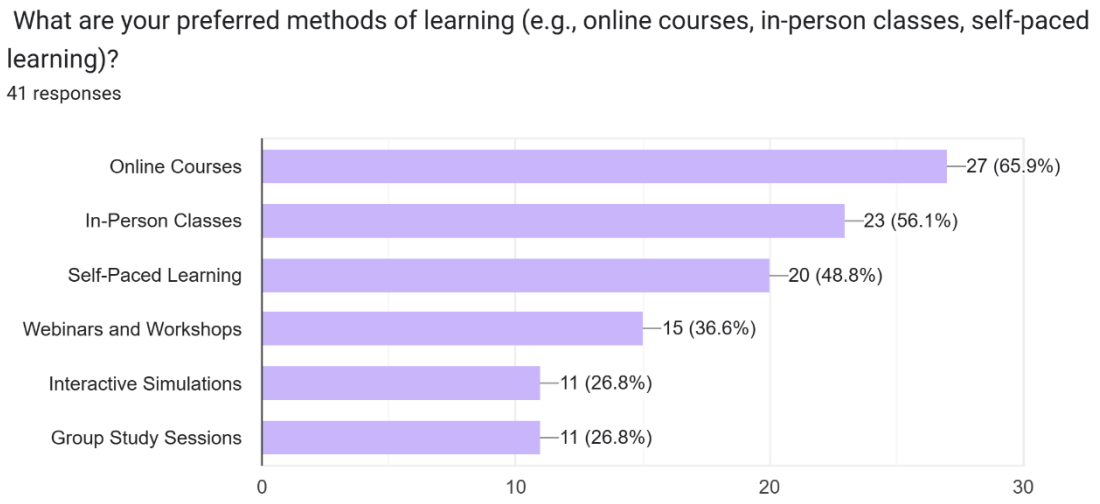


Figure 20: Responses of question 11.

The most preferred methods for learners are by enrolling on Online Courses where online courses can bring many benefits and convenience to the learners. The second and third preferred will be in-person classes and Self-paced Learning as both can bring a more personalized experience to maximize the efficiency of the learning path. The least preferred methods are interactive simulations and group study sessions as they depend on different individuals and both interactive and self-focused sections will be equally important in the learning progress.

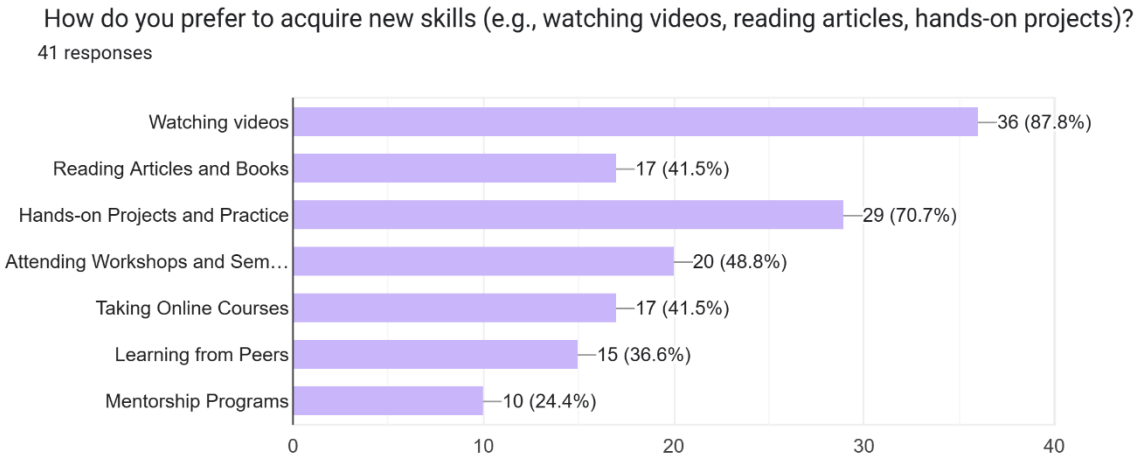


Figure 21: Responses of question 12.

36 Respondents (87.8%) prefer watching videos to acquire new skills and follow up with working on hands-on projects and practices with 29 respondents (70.7%). The least preferred way is with Mentorship programs which might be more costly compared to watching videos which might be free, but in terms of quality, they also has a big difference for the learners.

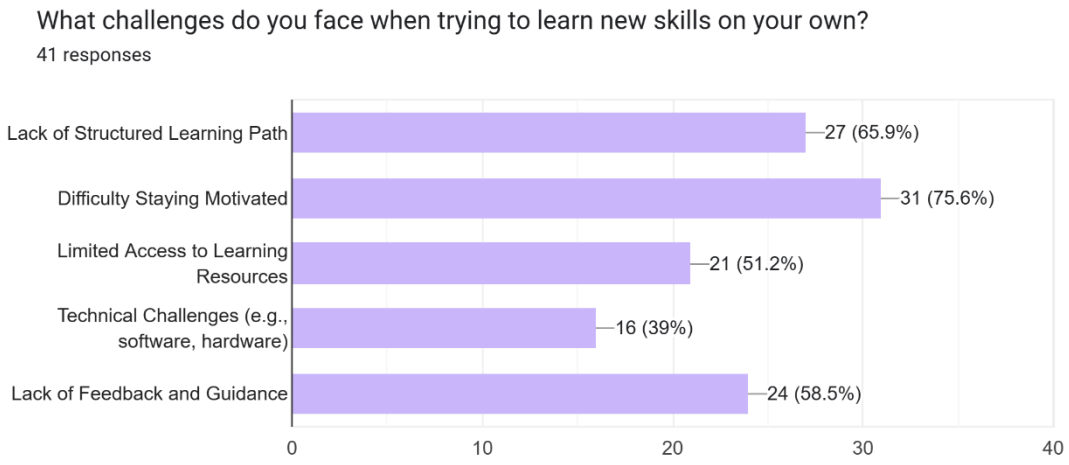


Figure 22: Responses of question 13.

The biggest challenges for the respondents is to staying in motivated when learning with 31 respondents (75.6%) , followed by Lack of structured learning path 27 respondents (65.9%) and lack of feedback and guidance 24 respondents (58.5%). These are the issue encountered and proposed solution by researcher need to aligned with the issue encountered.

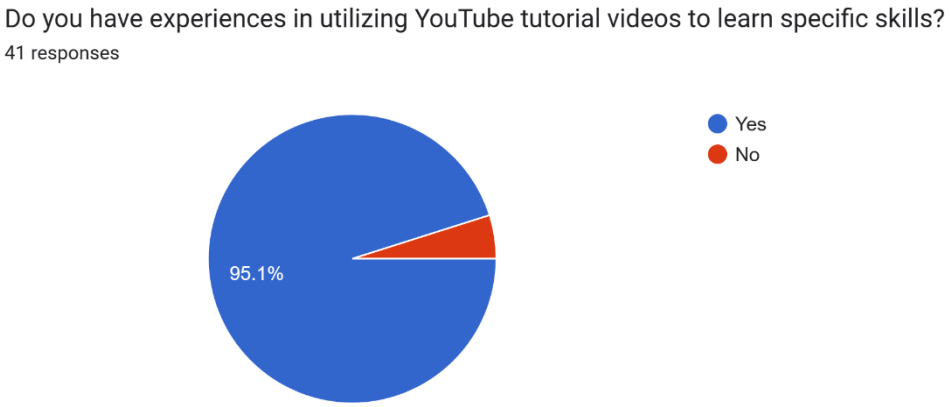


Figure 23: Responses of question 14.

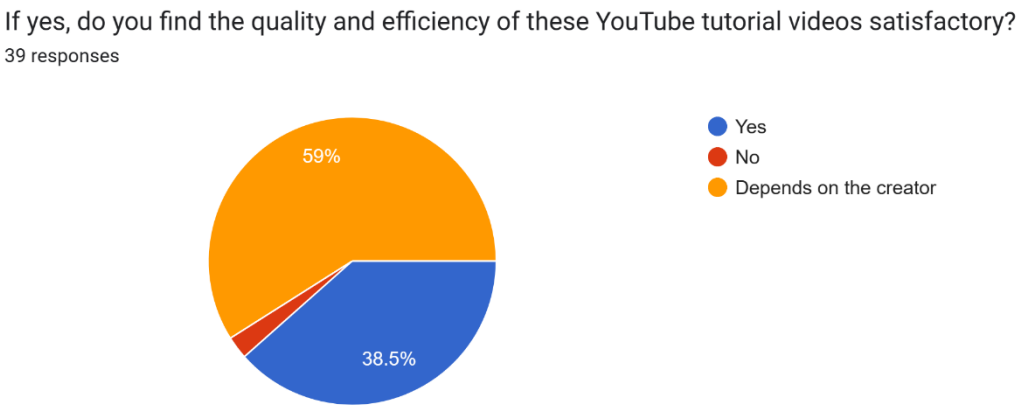


Figure 24: Responses of question 15.

From questions 14 and 15, most of the respondents 95.1% had experiences of watching YouTube videos to learn a certain skill. However, 59% of the respondents found that the quality varies with the creator of the content. 1 Respondent says these videos do not satisfy the needs of their learning expectations, and the rest about 38.5% are satisfied with the quality of these videos. From these responses, the certification and credibility of the creator have a great impact on the satisfaction of the users that want to learn from their videos.

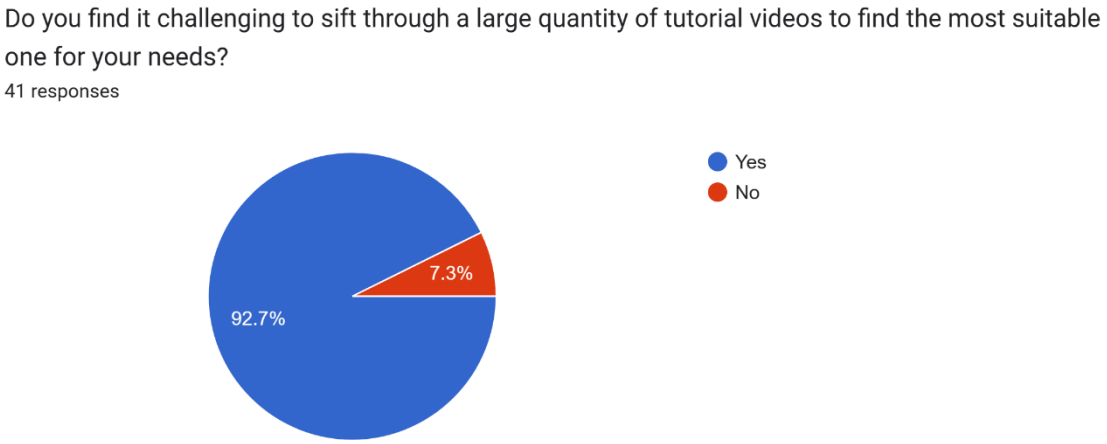


Figure 25: Responses of question 16.

About 92.7% of respondents find it challenging to sift through a large number of videos to find a suitable video. As from question 10, X respondents state that the relevance and suitability of course or material is the most important in online learning.

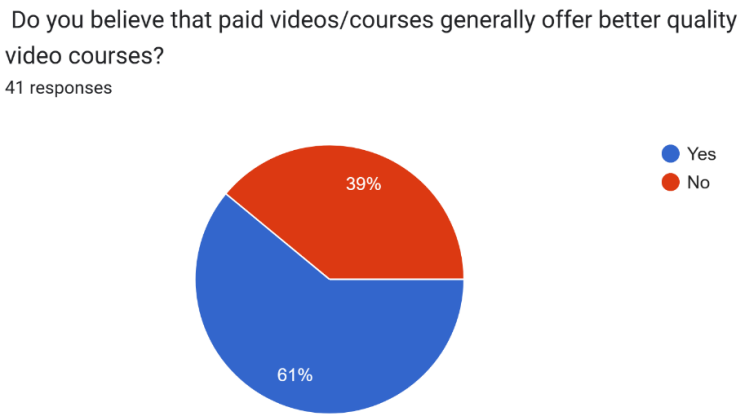


Figure 26: Responses of question 17.

In previous questions on getting preference on online courses, many stated the preferred budget would be RM200 and below or for free. However, 61% of respondents believe paid video courses are better quality generally compared to free courses. 39% of respondents do not agree that paid courses are generally better than free courses. Hence, both paid and free courses still have quality gaps but other elements such as relevance and reliability of courses have more impact on the experience of learners towards the courses.

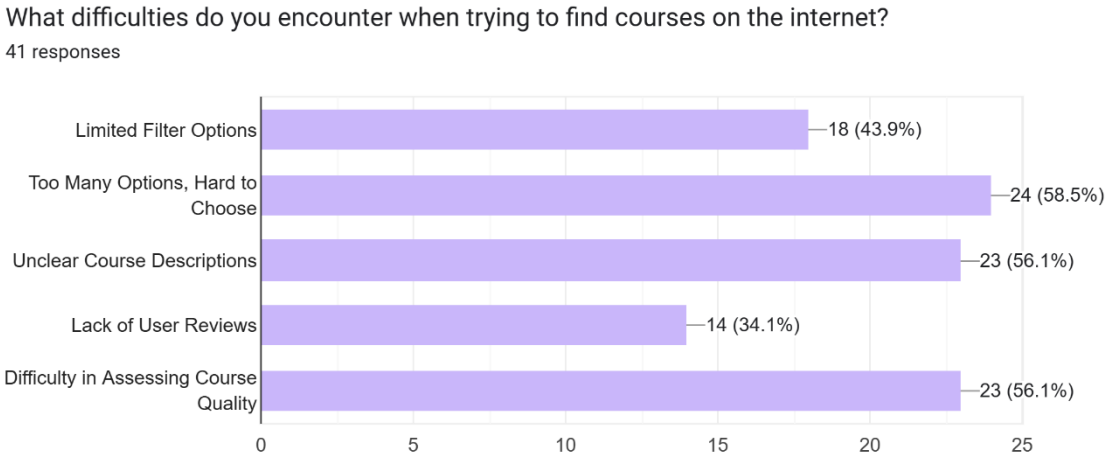


Figure 27: Responses of question 18.

The most found difficulties encountered in finding courses on the internet are Too many options lead to difficulties on selection, 24 respondents (58.5%). Unclear course descriptions and difficulties in assessing course quality, both having 23 respondents (56.1%) found they have these issues. These issues are related to the learner's efficiency in getting suitable courses for them. As having more content but the descriptions of the courses that are not standardised

and complete will make it difficulties for the learners to compare and decide which course to jump into without any additional tools available or asking from the experienced.

Summary for Analysis

In a nutshell, the main issue the learners encounter when seeking for online course is difficulty in finding the most suitable course for them with a large amount of choices with missing some information they hoped to know. Without adequate information and help, they are not able to decide which suits them bests which will result in giving up on learning that field from the platform. Although they are comfortable learning using YouTube videos, challenges such as difficulty in staying motivated and having a clear learning path to learn the skill they want are two of the important challenges during their learning process.

In terms of courses, most respondents tend to select the course that suits their budget below RM200, the creator is reliable, and the course content is relevant to the knowledge they want to learn. Some suggest that they need help in finding the course's name because they might not know the actual name but just an idea of what skill is used and related to which domain. Course with free preview is recommended and they should list all the information that the learner wants to know because they need to guarantee that the course Is what they are looking for.

Hence, the AI chatbot is proposed to solve and meet the issue and requirements to provide learners with a better experience using online platforms to learn their preferred skills.

User requirements of chatbot:

- It analyses the user preferences and produces recommendations for courses from different platforms such as Udemy and Coursera with budget constraints, creator's reliability, content relevance, updated contents, and preview availability applied.
- Provide course name-searching assistance for users who have only a general idea of the skill they want to learn.
- User-friendly interface of the system.
- Provide learning path guidance and tips and tricks by other users to stay motivated during the learning process.

3.5 Summary

In summary, research for physical and software requirements, development methodology, and data collection using survey and analysis are done in this chapter. The final user requirements of chatbots are produced depending on the issues and requirements collected from the respondents who participated.

CHAPTER 4: CONCLUSION

4.1 Achievements Reached

In the first part of the project, the aim and objective are drafted to solve problems addressed in the problem statement. Then, the researcher is knowledgeable on all the domains related to the AI, NLP, Chatbot and course learning platform and its course-seeking tool. This research has aided in aligning the problem statements and using the tools that are up-to-date in the industry. Technical concepts used are discussed and the tools needed are compared and justify the choice of tools to be used in the development phase of the project. The details of every process in the project are discussed in the methodology design section where each task and deliverables is clearly stated. The actual requirements of the chatbot are produced and added with some of the requirements stated by the respondents who participated in our survey.

4.2 Adequate Research and Analysis Justification

This project aims to achieve to have research on the developments of AI, and NLP. ML and propose a chatbot system development method and tools needed to complete the system. All the tools and software hardware requirements are listed. Ideas from the public are collected and analysed to find the requirements of the system. Hence, all of the investigation and research are done enough for the project as the objectives are met.

4.3 Gaps and Future Enhancement

There are several gaps in the research and the proposed system. Research on deploying edu-chatbot has no further deployment result data where the researcher can know the effectiveness of using chatbots to aid in the learning process compared to the traditional way. The dataset enquiry should have standards on giving open-source API to retrieve the data for their domain. It is because when making an aspect system chatbot, a new dataset need to be prepared and trained while when each of the school has produced these chatbots using different datasets, there will be a huge waste of resources to duplicate the training work.

The system could implement multiple input types for users such as attaching files, using speech-to-text, and integrating with the calendar and other scheduling applications to help in scheduling the learning path for the user. Chat history could be saved by each of the user’s accounts with integration using cloud storage to save cost on buying big server storage devices.

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APPENDICES

Screenshot of PPF

Office Record	Receipt
Date Received: Received by whom:	Student name: Student number: Received by: Date:

DRAFT PROJECT PROPOSAL FORM

Proposal ID:

Supervisor: **: Mr. AMAD ARSHAD**

Student Name **: LOO HOENG MENG**

Student No **: TP064158**

Email Address **:** TP064158@mail.apu.edu.my loo.hoengmeng@hotmail.com

Programme Name: Bachelors in computer science

Title of project **: Automated Course Seeking AI-Chatbot using Natural Language Processing (NLP)**

Please record which module(s) your topic is related to:

Introduction to Artificial Intelligence (112021-VTK)

Data Structures (112022-CMM)

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1. Introduction

Assume the reader has very little knowledge of the subject. Introduce the topic, the sector of business/industry concerned and how the project relates to it. Define the context of the problem and identify the research required to solve it.

In the modern era, online learning is a new trend because knowledge and information are easily searchable through the internet. They may come in different virtual forms: traditionally, such as digital news, news streaming platforms, articles, digital journeys, and research articles, to less formal types of information acquisition, such as TikTok and YouTube videos, and educational creators on Instagram. People can gain knowledge with a single touch of their fingers using the internet and their smartphones. However, variations in the quantity of knowledge on the internet could raise many issues related to the quality, relevance, and integrity of the information when one is serious about learning a certain skill set.

Figure 28:PPF page 1

Artificial Intelligence (AI) has been around for quite some time; many call them smart machines that can perform tasks that require the intelligence of humans. According to Copeland (2023), AI is the ability of digital computers or robots to perform tasks commonly associated with intelligent beings. Most modern AI chatbots can comprehend human input and reply in a natural way of speaking. This is achieved using the technology of Natural Language Processing (NLP), considered a branch of AI that helps computers understand, interpret, and manipulate human language (Natural Language Processing (NLP): What It Is and Why It Matters, n.d.). AI chatbots with NLP can provide decent responses depending on user input and perform some actions using predefined task automation. Automation refers to automating workflows on the computer as well as using online tools to execute instant commands. AI chatbots with automation are very convenient and powerful tools for helping with daily human tasks.

As the rise of Artificial Intelligence adopts the use of natural language processing (NLP), powerful AI tools are emerging to help humans in their daily lives by utilizing NLP and training models with big data techniques to acquire more accurate data in a more human-like manner, as seen in tools like ChatGPT and Barn by Google. Recently, it has become a norm for society to use AI after the launch of ChatGPT by OpenAI on November 30, 2022 (OpenAI, 2022), as a helper in their work, providing guidance and ideas after receiving prompts from users.

The automated course-seeking AI chatbot using NLP is a research project aimed at developing an AI chatbot that can help students find the most suitable courses for their needs and budget. As the types and prices of courses vary, this chatbot captures all the data and analyses it for the students. Through NLP, the chatbot can understand the student's preferences and requirements, providing suitable solutions such as courses for them to achieve their needs. A processed dataset from Kaggle related to Udemy course data is being used in this system. However, data cleaning and processing will not be the focus of this project.

Figure 29:PPF page 2

2. Problem Statement

Identify past and current work in the subject area. Outline the key references to other people's work, indicate for the most pertinent of these how your proposal relates to the ideas they contain.

1. Overwhelming number of courses available on online learning platforms.

Nowadays, the quality of courses may not be the same as the quality of courses on most online learning platforms. This will add difficulty for people who want to find the most suitable course for themselves. The reason is that there are many platforms such as Udemy or FreeCodeCamps (Perry, 2023) that provide the same course but from different creators, different levels of tutorials as well as teaching styles and lengths of tutorial. That will be time-wasting and money-wasting for those who have spent their time and money to learn something that is not worth or relevant. For people who have a limited budget for learning, free tutorials are the best options for them. However, according to research by Clifton & Mann (2011), content on free sources such as YouTube tends to be misleading, inaccurate, and biased. Learners need to have the ability to identify which and where to find the best learning courses on the internet.

2. Many learners, particularly pre-university students, often have little or no ideas about what they want to learn.

For secondary fresh graduates, they reach a point where they must decide on their academic path. Some may take a break before continuing their studies, others may follow their parents' decisions, and some may directly enter the workforce. Regardless, they need to explore their interests by gaining knowledge in different fields to see which one best suits them.

Some will have a brief understanding through YouTube videos, and some will directly purchase courses to try them out. Some tutorials are too lengthy and lack quality content, wasting their time and money. This is exacerbated when it comes to paid courses and videos. For those who directly enter college life, whether compelled or out of curiosity, standard courses in colleges and universities can take 3 to 4 years and cost hundreds of thousands of ringgits to complete a degree certification course. If students realize this is not the path they desire, they face the dilemma of either persisting through a stressful experience or dropping out midway, wasting both money and several years in what we call an exploration period. Many struggle in their further studies due to a lack of guidance and help.

Figure 30:PPF page 3

3. No course recommendation chatbot on the internet.

As stated above, many might struggle to select the most suitable courses online from various platforms. Furthermore, the search functions have many filters that might be troublesome for the learner who does not know a certain field. For example, students could be confused to differentiate between engineering, computer engineering and software engineering if they did not do further exploration on the internet. Reviews and comments are only available for the students to know the actual quality of the courses. However, this will only be trusted when the number of reviews is many, but not all the courses have thousands of students that are enrolled in any given review. New courses but having not many students to review might not be outstanding than the courses that have been enrolling for many years.

Figure 31:PPF page 4

3. Project Aims and Objectives

Identify the AIM(s) of the project, i.e. what the overall achievement is intended to be, in terms of both academic and commercial/industrial advances. Identify the particular intellectual difficulties posed by the proposal, the problems to be addressed, and explain how these might be solved. Clearly list individual measurable OBJECTIVES which can be related to the workplan and deliverables.

Aims and objectives are subject to approval from supervisor and students are expected to revise them if deemed inappropriate for a Level 3 project.

The project aims to develop an AI chatbot that can help students find the most suitable courses for their needs and budget.

The project's objectives are as follows:

- To explore the development of natural language processing integrated with AI chatbots and implementation on the market.
- To analyse similar systems that have AI chatbots using natural language processing to assist the user in finding their preference needs.
- To evaluate the chosen design and implementation approach in the development of NLP implemented AI chatbot.
- To collect the important preferences and requirements of students in course seeking online.
- To evaluate the chatbot's performance from multiple aspects such as reliability, user-friendly, speed, integrity and accuracy depending on user feedback.

Figure 32:PPF page 5

4. Literature Review

To further evaluate the topic "Automated Course Seeking AI-Chatbot using Natural Language Processing (NLP)", the researcher will explore the implementation of chatbots with natural language processing from the first chatbot to the modern usage of chatbots in society. The usage of natural language processing will be explored as well to have a brief idea of the usage of natural language processing and how chatbots and NLP integrate.

The first ever chatbot named Eliza was first introduced by Weizenbaum (1966) four decades ago. Eliza is a chatbot from a computer that can give a response upon user inputs that's first even made so computers and humans can have a conversation together. The concepts of the reply mechanism focused on how a computer can understand the prompt entered by the users, which are identification of keywords, discovery of minimal contexts, choices of appropriate transformations, generation of responses in the absence of keywords and finally provision of an editing capability for ELIZA's replying to scripts. These concepts formed the fundamental procedures of machine attempts to digest natural language, and they are being used in the natural language processing (NLP) development concepts.

Nowadays, chatbots are greatly advanced and they are being used in many fields, among the chatbots conversational chatbots are the most popular ones. These chatbots are equipped with Natural language processing to allow the computer to comprehend human languages and use deep learning techniques to allow responses with decent answers. For instance, a Xatkit development framework-based chatbot system architecture proposed by Baha et al. (2022) is capable of receiving string and audio inputs, processed by an in-house NLP engine to process intent detection and recognition. The chatbot's output is not a computer-generated plain text, but a 3D avatar can react with certain gestures, and mimics and continue the conversation with the audio playback of the generated responses. This chatbot is integrated with multiple modern technologies of NLP, deep learning, intent recognition, and Speech Synthesizer module to turn text into speeches that could sync with the gesture of the avatar. This Edu chatbot is implemented in the classrooms of schools.

Figure 33:PPF page 6

A systematic review of chatbot applications in the education field was carried out by Okonkwo and Ade-Ibijola (2021), they found many articles focused on chatbots are useful as a learning aid in teaching and learning such as personal learning, automation of administration work, aid in assessment, advising on educational issues, research and development aids. For the limitations, they found that chatbots have several issues with ethical problems of artificial, accurate evaluations, user attitude, chatbot supervision and difficulty in maintenance. In the end, the researchers proposed future research work for the research domain to further enhance the chatbots with technological advancements, completing the ethical principles developments of artificial tools so they will be safe to use in daily life as well as usability testing. As chatbots are further enhanced and grow, it is necessary to have well-defined rules for using chatbots at work so they will be compatible with user ethics.

According to research by Steele (2023), she has pointed out three major threats of ChatGPT to the students in education performance and added her counterpoints to those issues. The issues are the field of measurement of the student's actual ability, providing information inaccuracy and skill devaluation because the GPT can do a lot of writing for humans in just a few seconds. On the other hand, she emphasized harnessing the potential of these tools that can help the students to learn, as well as mitigate the threats, as the AI could also foster the critical thinking of the students when they use the ChatGPT to comprehend something. They give up rapidly. In the end, this researcher thinks that we must make use of these tools to forge the future.

These powerful chatbots have a primary ability to have before performing the tasks, that is to comprehend the needs of the human input and give the best right answer. The NLP is the core technology that is used to tell the computer what to do with human sentences. NLP is being used in other fields such as user posting on social media platforms. According to research by Valdez et al. (2023b), they have used NLP to analyze tweet postings on X, formerly named Tweeter, about vaccines for misinformation indicators during the COVID-19 times. The NLP analyses support two different languages English and Spanish. In conclusion, they found misinformation in both languages, and some of them overlap and they have provided misinformation in different areas related to vaccination distribution. This is one of the implementations of NLP used for analysing human posting sentences.

Figure 34:PPF page 7

Natural language processing has data processing phrases certain techniques are used after tokenizing the sentences for the computer to process. The traditional ways of processing tokenized words are by logistic regression. (DeepLearning.AI, 2023) However, Chatbots such as Google's Barn and ChatGPT have huge NLP models that are commonly called large language models (LLM) to generate texts that are accurate for commercial application use. These LLMs require high computational power and memory that will learn to infrastructure cost. (Barla, 2023) Quantum machines are proposed for use in NLP applications as quantum machines provide the capability of solving problems that normal computers are difficult to solve by using quantum mechanics. Research by Pandey et al. (2023) combined quantum machines and machine learning techniques to compare the performance of using classical computing with ML for parts of speech tagging of code-mixed datasets. They found out that applying quantum to long short-term memory will enhance the performance of POS tagging of social media code-mixed language on multiple platforms. Those algorithms used in quantum machine learning are beneficial to natural language processing applications.

The researchers have shown the application of NLP for different uses, however, chatbots are machines that use NLP together with artificial intelligence that can improve accuracy and do much more. According to research by Dongbo et al. (2023), they pointed out that many pieces of research related to chatbots have one issue, that is those chatbots do not accurately identify the user's requirements. Hence, they have proposed a chatbots using Bi-directional recurrent Neural Network with a Fuzzy Naïve Bayes classifier(BRNN-FNB) that helps in identifying the requirements and an AI-based sentimental analysis to perform accurate interactions. The result of the research is they got a 93% accuracy of the proposed chatbots with BRNN-FNB using the seq-to-seq techniques, and 92% per cent of accurate without seq-to-seq techniques. The seq-to-seq model is a class of Recurrent Neural Network architecture that is used to solve complex language problems such as creating chatbots.

Chatbots are useful to implement in business intelligence systems(BIS) as well. According to Azmi et al. (2023), they have done case studies on AI with BISs where AI brings many powerful and convenient capabilities to the systems. The capabilities such as enhancing data analysis and providing predictive analysis as well, as automating data integrations by

Figure 35:PPF page 8

providing natural language interfaces, detecting anomalies in the operations, enhancing data visualization into charts and diagrams and delivering personalized recommendations for users. They proposed a flowchart on the business information system, and where and what AI capabilities can be used to ease the operation of BIS and increase the accuracy as well.

The technology, concepts and implementation of chatbots and natural language processing from the first chatbot in 1966 to the modern advanced chatbots are discussed. Some issues and solutions of AI are pointed out. Some solutions are being implemented to fix the issues such as quantum machine learning to fix memory and computing power lacking problems. Examples of AI chatbot implementations in the educational field are being discussed. In conclusion, AI chatbot with NLP has great potential to enhance the quality of work for human society. Although technical and ethical issues arise when one breakthrough is made, the chatbots are nevertheless helped in human society in many ways. In this project, we will implement the AI chatbot to help in solving SDG 8, to provide sustainable work opportunities for everyone by allowing them to have the best learning opportunities with the help of technology.

5. Deliverables

Provide a clear list of the outputs from the project.

The system is a working prototype of the AI chatbot. It could respond with the related course names for the students after getting the preferences of the students through the data collection process when chatting with the users.

- It allows users to tell the chatbot their preferences and capture them in sentence form.
- It allows users to give further information about their preferences if the data acquired is unclear or missing.
- It allows users to connect with live chat with real people when help is needed.
- It provides a user manual and demo videos are available upon request to the chatbot.
- It enables users to get the most relevant course from the course platform.

Figure 36:PPF page 9

6. References

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Figure 38:PPF page 11

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Figure 39:PPF page 12

Screenshot of signed ethics form

Office Record	Receipt – Fast-Track Ethical Approval
Date Received:	Student name: Loo Hoeng Meng Student number: TP064158
Received by whom:	Received by: Date:

APU / APIIT FAST-TRACK ETHICAL APPROVAL FORM (STUDENTS)

Tick one box (level of study):
- ☐ POSTGRADUATE (PhD / MPhil / Masters)
- √ UNDERGRADUATE (Bachelors degree)
- ☐ FOUNDATION / DIPLOMA / Other categories

Tick one box (purpose of approval):
- √Thesis / Dissertation / FYP project
- ☐ Module assignment
- ☐ Other: _____

Title of Programme on which enrolled Bachelor Degree in Computer Science.

Tick one box: √ Full-Time Study or ☐ Part-Time Study

Title of project / assignment: Automated Course Seeking AI-Chatbot using Natural Language Processing (NLP)

Name of student researcher: Loo Hoeng Meng

Student Researchers- please note that certain professional organisations have ethical guidelines that you may need to consult when completing this form.

Supervisors/Module Lecturers - please seek guidance from the Chair of the APU Research Ethics Committee if you are uncertain about any ethical issue arising from this application.

		YES	NO	N/A
1	Will you describe the main procedures to participants in advance, so that they are informed about what to expect?			√
2	Will you tell participants that their participation is voluntary?	√		
3	Will you obtain written consent for participation?	√		
4	If the research is observational, will you ask participants for their consent to being observed?			√
5	Will you tell participants that they may withdraw from the research at any time and for any reason?	√		
6	With questionnaires and interviews will you give participants the option of omitting questions they do not want to answer?	√		
7	Will you tell participants that their data will be treated with full confidentiality and that, if published, it will not be identifiable as theirs?	√		
8	Will you give participants the opportunity to be debriefed i.e. to find out more about the study and its results?	√		

If you have ticked **No** to any of Q1-8 you should complete the full Ethics Approval Form.

		YES	NO	N/A
9	Will your project/assignment deliberately mislead participants in any way?		√	
10	Is there any realistic risk of any participants experiencing either physical or psychological distress or discomfort?		√	
11	Is the nature of the research such that contentious or sensitive issues might be involved?		√	

If you have ticked **Yes** to 9, 10 or 11 you should complete the full Ethics Approval Form. In relation to question 10 this should include details of what you will tell participants to do if they should experience any problems (e.g. who they can contact for help). You may also need to consider risk assessment issues.

Fast-Track Ethical Approval Form ver. 3.0 (Dec 2015) Page 1 of 4

Figure 40: Signed ethics form page 1

			YES	NO	N/A
12	Does your project/assignment involve work with animals?			√	
13	Do participants fall into any of the following special groups? **Note that you may also need to obtain satisfactory clearance from the relevant authorities**	Children (under 18 years of age)		√	
		People with communication or learning difficulties			
		Patients			
		People in custody			
		People who could be regarded as vulnerable			
		People engaged in illegal activities (eg drug taking)			
14	Does the project/assignment involve external funding or external collaboration where the funding body or external collaborative partner requires the University to provide evidence that the project/assignment had been subject to ethical scrutiny?			√	

If you have ticked **Yes** to 12, 13 or 14 you should complete the full Ethics Approval Form. There is an obligation on student and supervisor to bring to the attention of the APU Research Ethics Committee any issues with ethical implications not clearly covered by the above checklist.

STUDENT RESEARCHER

Provide in the boxes below (plus any other appended details) information required in support of your application, THEN SIGN THE FORM.

	Please Tick Boxes
I consider that this project/assignment has no significant ethical implications requiring a full ethics submission to the APU Research Ethics Committee.	√
Give a brief description of participants and procedure (methods, tests used etc) in up to 150 words. This project aims to create a Course Recommendation System with an AI-chatbot equipped with natural language processing (NLP). The development process follows the Waterfall Technique, including surveys conducted during the planning and design stages. The study involves 50 adults aged 18 and above, gathering insights into their tutorial preferences, self-learning frequency, and the importance of an AI-chatbot in finding suitable courses. Data collection takes a month, followed by a two-week analysis period. The system undergoes multiple testing, including unit, integration, and system tests, before being made available to users. The goal is to ensure the AI-chatbot aligns with user needs, providing effective and personalized course recommendations.	
I also confirm that: i) All key documents e.g. consent form, information sheet, questionnaire/interview are appended to this application. Or	√
ii) Any key documents e.g. consent form, information sheet, questionnaire/interview schedules which need to be finalised following initial investigations will be submitted for approval by the project/assignment supervisor/module lecturer before they are used in primary data collection.	

E-signature: *Hoeng.Meng* ... ... Print Name: Loo Hoeng Meng ... ... Date: 10/11/2023
(Student Researcher)

Fast-Track Ethical Approval Form ver. 3.0 (Dec 2015) Page 2 of 4

Figure 41: Signed ethics form page 2

Please note that any variation to that contained within this document that in any way affects ethical issues of the stated research requires the appending of new ethical details. New ethical consent may need to be sought.
The completed form (and any attachments) should be submitted for consideration by your Supervisor/Module Lecturer

SUPERVISOR/MODULE LECTURER
PLEASE CONFIRM THE FOLLOWING:

	Please Tick Box
I consider that this project/assignment has no significant ethical implications requiring a full ethics submission to the APU Research Ethics Committee	√
i) I have checked and approved the key documents required for this proposal (e.g. consent form, information sheet, questionnaire, interview schedule) Or	√
ii) I have checked and approved draft documents required for this proposal which provide a basis for the preliminary investigations which will inform the main research study. I have informed the student researcher that finalised and additional documents (e.g. consent form, information sheet, questionnaire, interview schedule) must be submitted for approval by me before they are used for primary data collection.	

SUPERVISOR AND SECOND ACADEMIC SIGNATORY

STATEMENT OF ETHICAL APPROVAL (please delete as appropriate)

1) **THIS PROJECT/ASSIGNMENT HAS BEEN CONSIDERED USING AGREED APU/SU PROCEDURES AND IS NOW APPROVED**

2) **THIS PROJECT/ASSIGNMENT HAS BEEN APPROVED IN PRINCIPLE AS INVOLVING NO SIGNIFICANT ETHICAL IMPLICATIONS, BUT FINAL APPROVAL FOR DATA COLLECTION IS SUBJECT TO THE SUBMISSION OF KEY DOCUMENTS FOR APPROVAL BY SUPERVISOR (see Appendix A)**

E-signature... ... ... Print Name: Mr Amad Arshad... ... Date...13/11/2023 ...
(Supervisor/Lecturer)

E-signature... ... ... Print Name: Mr. Lee Kim Keong ... ... Date... ... ...
(Second Academic Signatory)

Fast-Track Ethical Approval Form ver. 3.0 (Dec 2015) Page 3 of 4

Figure 42: Signed ethics form page 3

Office Record	Receipt – Appendix A (Fast-Track Ethics Form)
Date Received:	Student name: Loo Hoeng Meng Student number: TP064158
Received by whom:	Received by: Date:

APPENDIX A
AUTHORISATION FOR USE OF KEY DOCUMENTS

Completion of Appendix A is required when for good reasons key documents are not available when a fast track application is approved by the supervisor/module lecturer and second academic signatory.

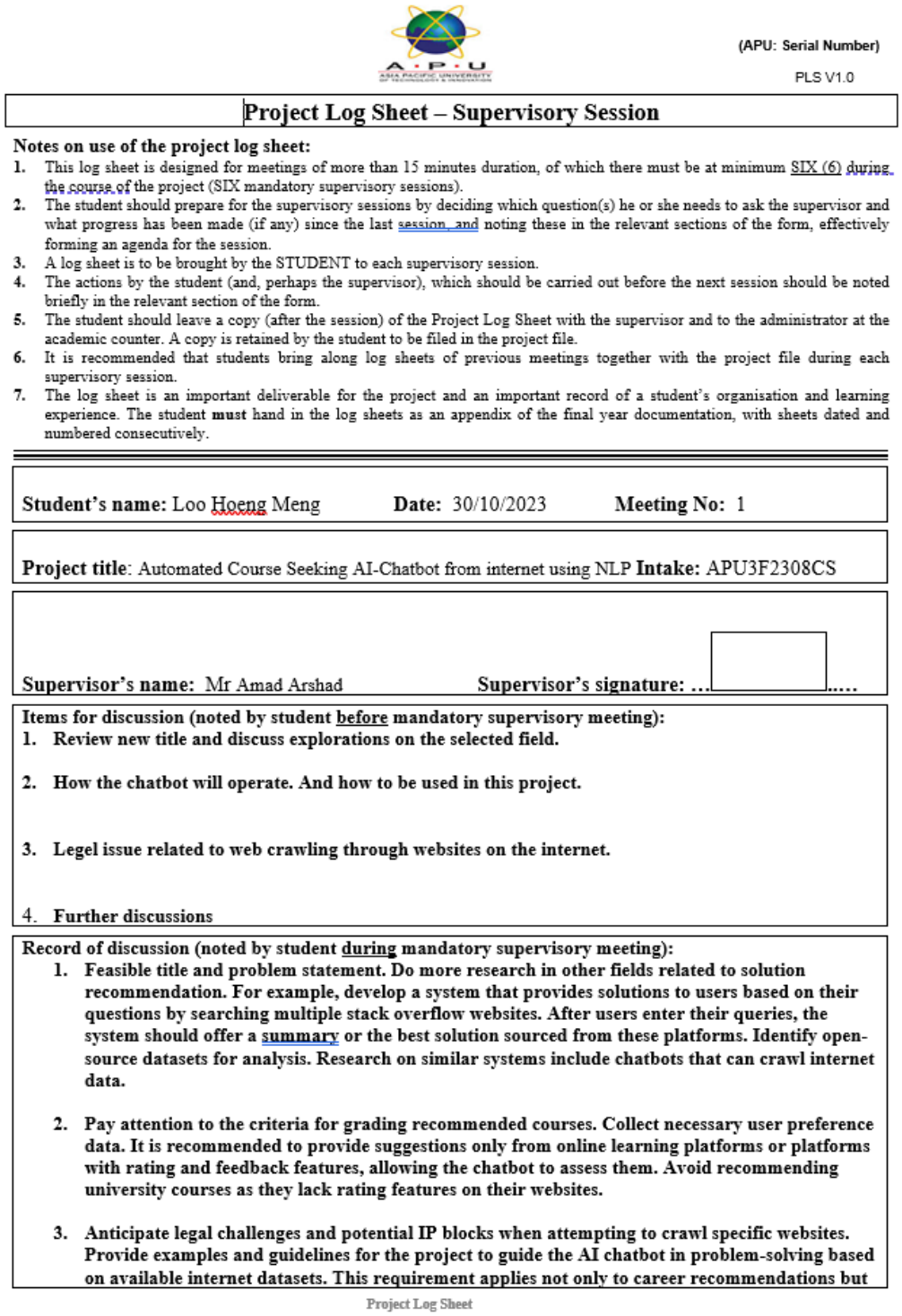
I have now checked and approved all the key documents associated with this proposal e.g. consent form, information sheet, questionnaire, interview schedule

Title of project/assignment: … … … **Automated Course Seeking AI-Chatbot using Natural Language Processing (NLP)**

… … … …

Screenshot of Log sheets

Log sheets 1



(APU: Serial Number)

PLS V1.0

Project Log Sheet – Supervisory Session

Notes on use of the project log sheet:

1. This log sheet is designed for meetings of more than 15 minutes duration, of which there must be at minimum SIX (6) during the course of the project (SIX mandatory supervisory sessions).
2. The student should prepare for the supervisory sessions by deciding which question(s) he or she needs to ask the supervisor and what progress has been made (if any) since the last session, and noting these in the relevant sections of the form, effectively forming an agenda for the session.
3. A log sheet is to be brought by the STUDENT to each supervisory session.
4. The actions by the student (and, perhaps the supervisor), which should be carried out before the next session should be noted briefly in the relevant section of the form.
5. The student should leave a copy (after the session) of the Project Log Sheet with the supervisor and to the administrator at the academic counter. A copy is retained by the student to be filed in the project file.
6. It is recommended that students bring along log sheets of previous meetings together with the project file during each supervisory session.
7. The log sheet is an important deliverable for the project and an important record of a student's organisation and learning experience. The student **must** hand in the log sheets as an appendix of the final year documentation, with sheets dated and numbered consecutively.

Student's name: Loo Hoeng Meng **Date:** 30/10/2023 **Meeting No:** 1

Project title: Automated Course Seeking AI-Chatbot from internet using NLP **Intake:** APU3F2308CS

Supervisor's name: Mr Amad Arshad **Supervisor's signature:** ……

Items for discussion (noted by student before mandatory supervisory meeting):

1. **Review new title and discuss explorations on the selected field.**
2. **How the chatbot will operate. And how to be used in this project.**
3. **Legel issue related to web crawling through websites on the internet.**
4. **Further discussions**

Record of discussion (noted by student during mandatory supervisory meeting):

1. **Feasible title and problem statement. Do more research in other fields related to solution recommendation. For example, develop a system that provides solutions to users based on their questions by searching multiple stack overflow websites. After users enter their queries, the system should offer a summary or the best solution sourced from these platforms. Identify open-source datasets for analysis. Research on similar systems include chatbots that can crawl internet data.**
2. **Pay attention to the criteria for grading recommended courses. Collect necessary user preference data. It is recommended to provide suggestions only from online learning platforms or platforms with rating and feedback features, allowing the chatbot to assess them. Avoid recommending university courses as they lack rating features on their websites.**
3. **Anticipate legal challenges and potential IP blocks when attempting to crawl specific websites. Provide examples and guidelines for the project to guide the AI chatbot in problem-solving based on available internet datasets. This requirement applies not only to career recommendations but**

Project Log Sheet

Figure 44: Supervisory session log sheet 1 - page 1.

other areas as well. 4. Emphasize the significance of datasets for the chatbot's functionality. Explore third-party repositories for related open-source datasets or APIs. APIs are preferable, especially if they offer free sample data.
Action List (to be attempted or completed by student by the <u>next</u> mandatory supervisory meeting): 1. Conduct Research on Available Datasets: Research available datasets and formulate potential titles and required content for the PPF. Explore the following aspects: a. Identify accessible datasets. b. Verify permissions for datasets (open-source, API availability, or free sample usage). c. Investigate similar chatbots that crawl data from the internet. 2. Survey and Ethical Form Completion: Once a suitable field is confirmed, ensure timely completion of the survey and ethical forms before the specified deadline. 3. Develop Comprehensive Project Information: Enhance project details, including the title and other relevant information. Schedule a discussion with Mr. Amad to finalize these details by next week.

Note: A student should make an appointment to meet his or her supervisor (via the consultation system) at least ONE (1) week prior to a mandatory supervisor session – please see document on project timelines. In the event a supervisor could not be booked for consultation, the project manager should be informed ONE (1) week prior to the session so that a meeting can be subsequently arranged.

Project Log Sheet

Figure 45: Supervisory session log sheet 1 - page 2.

Log sheets 2

(APU: Serial Number)

PLS V1.0

Project Log Sheet – Supervisory Session

Notes on use of the project log sheet:

8. This log sheet is designed for meetings of more than 15 minutes duration, of which there must be at minimum SIX (6) during the course of the project (SIX mandatory supervisory sessions).
9. The student should prepare for the supervisory sessions by deciding which question(s) he or she needs to ask the supervisor and what progress has been made (if any) since the last session, and noting these in the relevant sections of the form, effectively forming an agenda for the session.
10. A log sheet is to be brought by the STUDENT to each supervisory session.
11. The actions by the student (and, perhaps the supervisor), which should be carried out before the next session should be noted briefly in the relevant section of the form.
12. The student should leave a copy (after the session) of the Project Log Sheet with the supervisor and to the administrator at the academic counter. A copy is retained by the student to be filed in the project file.
13. It is recommended that students bring along log sheets of previous meetings together with the project file during each supervisory session.
14. The log sheet is an important deliverable for the project and an important record of a student's organisation and learning experience. The student **must** hand in the log sheets as an appendix of the final year documentation, with sheets dated and numbered consecutively.

Student's name: Loo Hoeng Meng **Date:** 10/11/2023 **Meeting No:** 2

Project title: Automated Course Seeking AI-Chatbot using Natural Language Processing (NLP) **Intake:** APU3F2308CS

Supervisor's name: Mr Amad Arshad **Supervisor's signature:** ... ...

Items for discussion (noted by student before mandatory supervisory meeting):

1. **Questionnaire review**
2. **Investigation Research status update.**

Record of discussion (noted by student during mandatory supervisory meeting):

1. **Questions are being reviewed and modified. Some question need further refinement will be modified and send to Mr. Amad for signature.**
2. **Using processed dataset from Kaggle is ok for the chatbot. Use natural language toolkit from with Python ok. These research results need to include into the technical investigation section.**

Action List (to be attempted or completed by student by the next mandatory supervisory meeting):

4. **Send the ethic form with refined survey questions.**
5. **Start sending the survey questions after approval.**
6. **Send title adjustment email to Dr Dewi.**
7. **Start doing IR and refine PPF.**

Note: A student should make an appointment to meet his or her supervisor (via the consultation system) at least ONE (1) week prior to a mandatory supervisor session – please see document on project timelines. In the event a supervisor could not be booked for consultation, the project manager should be informed ONE (1) week prior to the session so that a meeting can be subsequently arranged.

Project Log Sheet

Figure 46: Supervisory session log sheet 2

Log sheets 3

(APU: Serial Number)

PLS V1.0

Project Log Sheet – Supervisory Session

Notes on use of the project log sheet:

1. This log sheet is designed for meetings of more than 15 minutes duration, of which there must be at minimum SIX (6) during the course of the project (SIX mandatory supervisory sessions).
2. The student should prepare for the supervisory sessions by deciding which question(s) he or she needs to ask the supervisor and what progress has been made (if any) since the last session, and noting these in the relevant sections of the form, effectively forming an agenda for the session.
3. A log sheet is to be brought by the STUDENT to each supervisory session.
4. The actions by the student (and, perhaps the supervisor), which should be carried out before the next session should be noted briefly in the relevant section of the form.
5. The student should leave a copy (after the session) of the Project Log Sheet with the supervisor and to the administrator at the academic counter. A copy is retained by the student to be filed in the project file.
6. It is recommended that students bring along log sheets of previous meetings together with the project file during each supervisory session.
7. The log sheet is an important deliverable for the project and an important record of a student's organisation and learning experience. The student **must** hand in the log sheets as an appendix of the final year documentation, with sheets dated and numbered consecutively.

Student's name: Loo Hoeng Meng **Date:** **22**/11/2023 **Meeting No:** 3

Project title: Automated Course Seeking AI-Chatbot using Natural Language Processing (NLP) **Intake:** APU3F2308CS

Supervisor's name: Mr Amad Arshad **Supervisor's signature:** …

Items for discussion (noted by student before mandatory supervisory meeting):

1. **Investigation Report Review.**

Record of discussion (noted by student during mandatory supervisory meeting):

1. **Grammar needs to be fixed. Need to use academic vocab in the IR. Refine the problem contexts.**

Action List (to be attempted or completed by student by the next mandatory supervisory meeting):

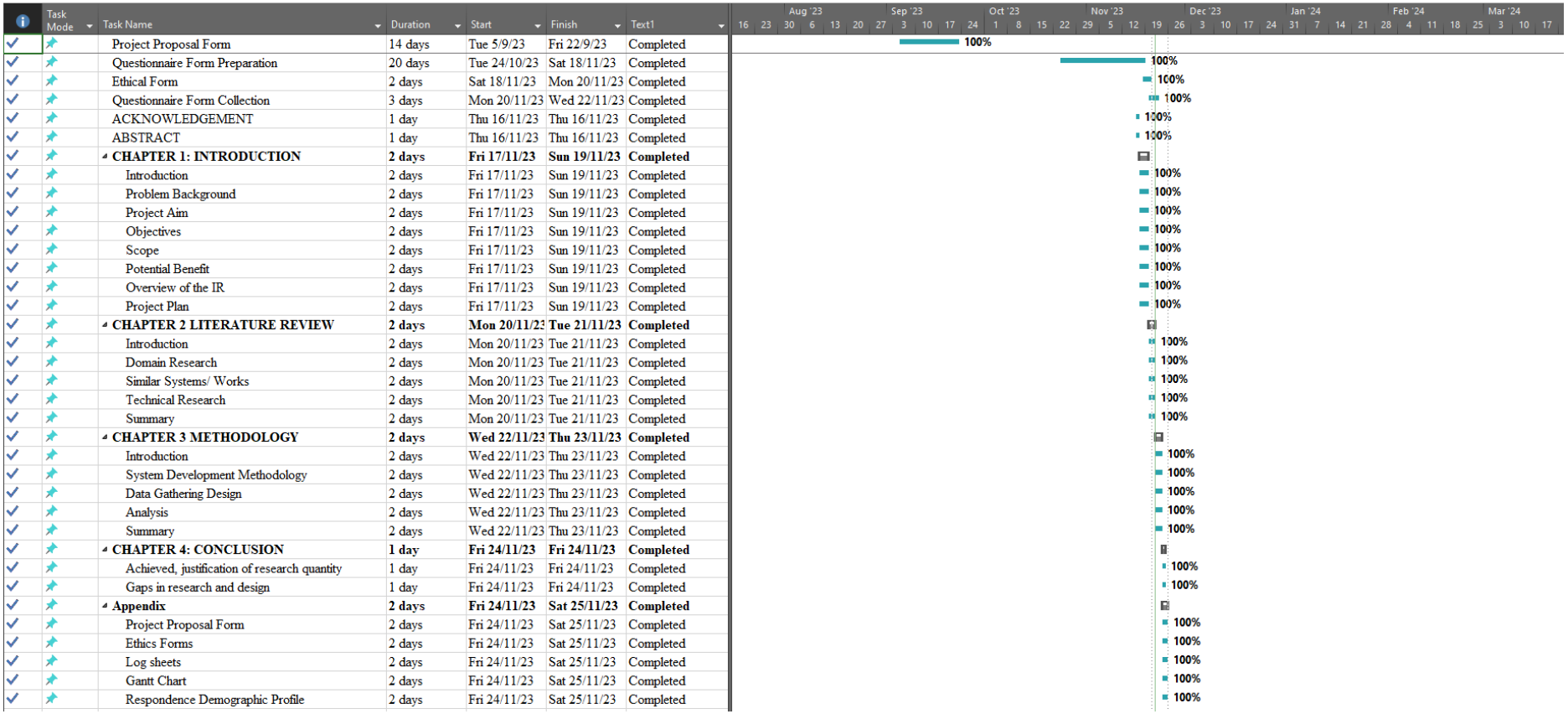
1. **Refine and add evaluation on the uncompleted parts. Schedule another meeting to further review the IR before the submission date.**

Note: A student should make an appointment to meet his or her supervisor (via the consultation system) at least ONE (1) week prior to a mandatory supervisor session – please see document on project timelines. In the event a supervisor could not be booked for consultation, the project manager should be informed ONE (1) week prior to the session so that a meeting can be subsequently arranged.

Project Log Sheet

Figure 47: Supervisory session log sheet 3

Gantt chart for FYP Semester 1



Task Mode	Task Name	Duration	Start	Finish	Text1
	Project Proposal Form	14 days	Tue 5/9/23	Fri 22/9/23	Completed
	Questionnaire Form Preparation	20 days	Tue 24/10/23	Sat 18/11/23	Completed
	Ethical Form	2 days	Sat 18/11/23	Mon 20/11/23	Completed
	Questionnaire Form Collection	3 days	Mon 20/11/23	Wed 22/11/23	Completed
	ACKNOWLEDGEMENT	1 day	Thu 16/11/23	Thu 16/11/23	Completed
	ABSTRACT	1 day	Thu 16/11/23	Thu 16/11/23	Completed
	CHAPTER 1: INTRODUCTION	**2 days**	**Fri 17/11/23**	**Sun 19/11/23**	**Completed**
	Introduction	2 days	Fri 17/11/23	Sun 19/11/23	Completed
	Problem Background	2 days	Fri 17/11/23	Sun 19/11/23	Completed
	Project Aim	2 days	Fri 17/11/23	Sun 19/11/23	Completed
	Objectives	2 days	Fri 17/11/23	Sun 19/11/23	Completed
	Scope	2 days	Fri 17/11/23	Sun 19/11/23	Completed
	Potential Benefit	2 days	Fri 17/11/23	Sun 19/11/23	Completed
	Overview of the IR	2 days	Fri 17/11/23	Sun 19/11/23	Completed
	Project Plan	2 days	Fri 17/11/23	Sun 19/11/23	Completed
	CHAPTER 2 LITERATURE REVIEW	**2 days**	**Mon 20/11/23**	**Tue 21/11/23**	**Completed**
	Introduction	2 days	Mon 20/11/23	Tue 21/11/23	Completed
	Domain Research	2 days	Mon 20/11/23	Tue 21/11/23	Completed
	Similar Systems/ Works	2 days	Mon 20/11/23	Tue 21/11/23	Completed
	Technical Research	2 days	Mon 20/11/23	Tue 21/11/23	Completed
	Summary	2 days	Mon 20/11/23	Tue 21/11/23	Completed
	CHAPTER 3 METHODOLOGY	**2 days**	**Wed 22/11/23**	**Thu 23/11/23**	**Completed**
	Introduction	2 days	Wed 22/11/23	Thu 23/11/23	Completed
	System Development Methodology	2 days	Wed 22/11/23	Thu 23/11/23	Completed
	Data Gathering Design	2 days	Wed 22/11/23	Thu 23/11/23	Completed
	Analysis	2 days	Wed 22/11/23	Thu 23/11/23	Completed
	Summary	2 days	Wed 22/11/23	Thu 23/11/23	Completed
	CHAPTER 4: CONCLUSION	**1 day**	**Fri 24/11/23**	**Fri 24/11/23**	**Completed**
	Achieved, justification of research quantity	1 day	Fri 24/11/23	Fri 24/11/23	Completed
	Gaps in research and design	1 day	Fri 24/11/23	Fri 24/11/23	Completed
	Appendix	**2 days**	**Fri 24/11/23**	**Sat 25/11/23**	**Completed**
	Project Proposal Form	2 days	Fri 24/11/23	Sat 25/11/23	Completed
	Ethics Forms	2 days	Fri 24/11/23	Sat 25/11/23	Completed
	Log sheets	2 days	Fri 24/11/23	Sat 25/11/23	Completed
	Gantt Chart	2 days	Fri 24/11/23	Sat 25/11/23	Completed
	Respondence Demographic Profile	2 days	Fri 24/11/23	Sat 25/11/23	Completed

Figure 48: Gantt Chart for FYP semester 1.

Respondence Demographic Profile

Table 6: Demographic profile table.

Nickname	Age Group	Nationality	Gender
Amanda	Under 20	Malaysian	Female
Ashley	Under 20	Malaysian	Male
Christopher	Under 20	Malaysian	Male
Daniel	Under 20	Malaysian	Male
David	Under 20	Malaysian	Male
James	Under 20	Malaysian	Male
Jennifer	Under 20	Malaysian	Female
Jessica	Under 20	Malaysian	Male
Kim	Under 20	Malaysian	Male
Matthew	Under 20	Malaysian	Male
Megan	Under 20	Malaysian	Female
Michael	Under 20	Malaysian	Male
Sarah	Under 20	Malaysian	Male
Joshua	Above 41	Malaysian	Female
John	31-40	Malaysian	Female
Lauren	31-40	Malaysian	Female
Amy	21-30	Malaysian	Male
Andrew	21-30	Malaysian	Male
Benjamin	21-30	Malaysian	Male
Brandon	21-30	Malaysian	Female
Brian	21-30	Malaysian	Male
Brittany	21-30	Malaysian	Male
Danielle	21-30	Malaysian	Male
Elizabeth	21-30	Malaysian	Male
Emily	21-30	Malaysian	Female
Heather	21-30	Malaysian	Female
Jason	21-30	Malaysian	Female
Joseph	21-30	Malaysian	Male
Justin	21-30	Malaysian	Male
Kimberly	21-30	Malaysian	Male
Laura	21-30	Malaysian	Male

Michelle	21-30	Malaysian	Male
Nathan	21-30	Malaysian	Female
Nicholas	21-30	Malaysian	Male
Nicole	21-30	Malaysian	Female
Rachel	21-30	Malaysian	Male
Robert	21-30	Malaysian	Male
Ryan	21-30	Malaysian	Female
Samantha	21-30	Malaysian	Male
Stephanie	21-30	Malaysian	Male
William	21-30	Malaysian	Male